

ming yin

THE DIVIDE-AND-CONQUER ALGORITHM FOR SYMMETRIC EIGENPROBLEMS

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THE DIVIDE-AND-CONQUER ALGORITHM FOR SYMMETRIC EIGENPROBLEMS

Ming Yin

Supervisor: Associate Professor William McLean

School of Mathematics and Statistics
UNSW Sydney

March 2026

SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS OF THE DEGREE OF
MASTER OF MATHEMATICS

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Abstract

Efficient eigensolvers for large dense symmetric matrices are critical in scientific computing, yet the practical difference between LAPACK's QR-based driver `DSYEV` and divide-and-conquer driver `DSYEVD` is not always clear. This thesis studies that difference from three complementary perspectives. First, it reviews the mathematical foundations of symmetric tridiagonal reduction and compares the QR-based and divide-and-conquer approaches for the tridiagonal eigenproblem. Second, it examines how these algorithmic differences appear in the implementation structure of LAPACK, especially in the routine paths built around `DSTEQR` and `DSTEDC`. Third, it reports controlled performance experiments on dense symmetric test matrices to evaluate how these structural differences affect runtime in practice. The results show that the main advantage of `DSYEVD` arises from algorithmic organization at the tridiagonal stage and from its more efficient use of blocked matrix operations, while SIMD provides only a smaller additional gain. In the tested setting, the two drivers behave similarly for eigenvalues-only workloads, whereas `DSYEVD` becomes substantially faster when eigenvectors are required, at the cost of higher workspace usage. These findings provide both a conceptual explanation and a practical guideline for selecting between the two routines.

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CHAPTER 1

Introduction

This thesis compares two algorithms for computing the eigenvalues and eigenvectors of dense symmetric matrices: the classical QR algorithm and the divide-and-conquer algorithm. It examines both their mathematical foundations and their implementation in LAPACK, through the drivers `DSYEV` and `DSYEVD`, and studies the practical computational consequences of their different algorithmic structures. More broadly, the thesis asks how these structural differences are reflected in theory, routine design, runtime behaviour, and practical routine selection.

1.1 Applications and Significance of the Eigenvalue Problem

The eigenvalue problem for real symmetric matrices is a fundamental task in numerical linear algebra. Owing to symmetry, such matrices admit real eigenvalues and an orthogonal basis of eigenvectors, making the problem both mathematically well-posed and numerically stable. Symmetric eigenvalue problems arise naturally in a wide range of applications, including vibration analysis, structural mechanics, and numerical simulations involving stability or resonance behavior. These problems frequently originate from finite difference or finite element discretizations, producing large symmetric matrices whose spectral properties influence stability and convergence.

The practical importance of symmetric eigenvalue problems is amplified by the scale of modern computations. Contemporary applications often involve matrices of large dimension, for which computational efficiency becomes critical. Classical methods such as the QR algorithm, combined with Householder tridiagonalization [1, 2], offer robustness but become prohibitive for large-scale problems. The divide-and-conquer approach, first proposed by Cuppen [3] and subsequently improved by Gu and Eisenstat [4], represents a significant advancement. This method decomposes the problem into smaller subproblems, solves them recursively, and exploits higher-level BLAS operations, making it particularly attractive for modern computing architectures [5].

This thesis aims to analyze the mathematical foundations and implementation details of the divide-and-conquer method as implemented in LAPACK's `DSYEVD` routine, compare its performance with the classical QR-based approach (`DSYEV`), and provide insights into practical algorithm selection for different problem sizes and computational environments.

1.2 LAPACK and BLAS as the Implementation Context

The Basic Linear Algebra Subprograms (BLAS) provide standard low-level kernels for vector and matrix operations, while LAPACK (Linear Algebra PACKage) builds

on these kernels to implement higher-level dense linear algebra algorithms [2]. In practice, LAPACK is the standard software library for dense numerical linear algebra, and its widespread use makes it a natural implementation context for the present study.

The distinction matters because LAPACK performance depends not only on the number of floating-point operations, but also on how effectively an algorithm can be organized around BLAS kernels, especially Level 3 BLAS matrix-matrix operations. Algorithms that expose more blocked linear algebra generally make better use of cache hierarchies and modern hardware, so differences in algorithmic structure can lead to substantial differences in practical runtime.

Within LAPACK, dense symmetric eigenvalue problems are handled by driver routines that follow a common overall structure: reduction of the dense matrix to tridiagonal form, solution of the tridiagonal eigenproblem, and, when required, back transformation of eigenvectors to the original dense problem. The routines `DSYEV` and `DSYEVD` share this framework but differ in the algorithm used at the tridiagonal stage, making them a natural pair for studying how mathematical structure, routine design, and measured performance fit together.

1.3 A Motivating Comparison of `DSYEV` and `DSYEVD`

The two routines solve the same dense symmetric eigenproblem and expose closely related interfaces, but they differ at the tridiagonal stage: `DSYEV` follows the classical QR-based path, whereas `DSYEVD` uses divide-and-conquer. When only eigenvalues are required, large performance differences are not generally expected; when both eigenvalues and eigenvectors are required, the divide-and-conquer path typically benefits more from blocked linear algebra and optimized BLAS kernels.

As an introductory preview, a small benchmark comparison was carried out for both the eigenvalues-only case and the full eigenpair case. The test uses four representative sizes aligned with the later Chapter 4 benchmark.

Table 1.1 summarizes wall-clock times (in seconds) and the speedup ordering implied by the ratio QR/DC (`DSYEV`/`DSYEVD`). Consistently, eigenvalues-only timings are nearly identical, whereas `DSYEVD` achieves substantial speedups in the full eigenpair case. In the aligned 512 to 4096 range, that speedup grows from about $4\times$ to about $5.4\times$.

This introductory comparison suggests two practical points. First, for eigenvalues only, the two routines behave very similarly. Second, when both eigenvalues and eigenvectors are required, `DSYEVD` is substantially faster in this benchmark. This supports the thesis focus on the tridiagonal eigensolver stage and on the implementation consequences of algorithmic structure. The table in this chapter is therefore used only as a motivating preview. A controlled experimental study is presented in Chapter 4.

Table 1.1: Timing comparison of DSYEV and DSYEVD.

Matrix size n	DSYEV (QR)		DSYEVD (D&C)	
	Eigvals [s]	Eigpairs [s]	Eigvals [s]	Eigpairs [s]
512	0.030	0.192	0.030	0.048
1024	0.230	1.427	0.232	0.323
2048	1.849	12.557	1.931	2.399
4096	17.294	106.606	17.208	19.804

1.4 Scope and Main Contributions

This thesis is limited to a focused comparison between DSYEV and DSYEVD, with particular emphasis on the tridiagonal eigensolver stage and its contribution to end-to-end runtime. It does not aim to survey the full range of dense symmetric eigensolvers or LAPACK drivers. Rather, it uses this comparison to relate mathematical structure, implementation design, and measured performance. SIMD is treated as a secondary implementation factor rather than as the main explanatory mechanism.

The main contributions are:

- A structured comparison of DSYEV and DSYEVD in terms of mathematical structure, LAPACK routine organization, and practical routine selection.
- A phase-level experimental analysis showing that performance differences are dominated by the tridiagonal eigensolver stage.
- A practical analysis of the runtime–workspace tradeoff between DSYEV and DSYEVD, showing that DSYEVD is preferable when both eigenvalues and eigenvectors are computed and runtime is the main concern, while DSYEV remains relevant in memory-constrained settings.

1.5 Organization of the Thesis

The remainder of the thesis is organized as follows. Chapter 2 develops the mathematical foundations needed for the comparison, including tridiagonalization, QR-based tridiagonal methods, and divide-and-conquer. Chapter 3 examines how these ideas are realized in LAPACK, with particular attention to the routine paths that distinguish DSYEV from DSYEVD. Chapter 4 presents the experimental study and interprets the measured performance differences through benchmark, profiling, and hardware-efficiency evidence. Chapter 5 draws together the main findings, practical guidance, limitations, and directions for future work.

CHAPTER 2

Mathematical Foundations of syev and syevd

This chapter reviews the core mathematics for solving real symmetric eigenproblems. The classical pipeline first applies orthogonal similarities to reduce a dense matrix to tridiagonal form, then uses a structure-preserving iteration on the tridiagonal matrix, and for large problems may switch to a divide-and-conquer merge.

Householder reflectors and Givens rotations are used throughout this chapter, but their detailed definitions are given within the algorithm-specific sections: **SYTRD** (Section 2.1) and **STEQR** (Section 2.2).

Section 2.1 revisits Householder tridiagonalization as the common first stage. Section 2.2 presents implicit QR/QL iteration for symmetric tridiagonal matrices. Section 2.3 outlines the divide-and-conquer strategy that merges subproblems through a rank-one correction. Section 2.4 then summarizes the conceptual, computational, and stability differences between these two tridiagonal eigensolvers, providing the bridge to the LAPACK implementation discussion in Chapter 3.

2.1 SYTRD: Householder Tridiagonalization

Given a real symmetric matrix $A \in \mathbb{R}^{n \times n}$, the goal is to construct an orthogonal matrix Q such that

$$T = Q^T A Q$$

is symmetric and tridiagonal. Since orthogonal similarity preserves eigenvalues, this reduction transforms the symmetric eigenvalue problem into a tridiagonal one without altering its spectrum. This method traces back to Householder and was formalised for the symmetric case by Martin–Reinsch–Wilkinson (1968). We write $A^{(j)}$ for the matrix after the j -th similarity step and use indices i for rows/columns; this avoids mixing iteration counters with position indices.

2.1.1 Householder reflector

Definition 2.1.1 (Householder reflector). *For a vector $x \in \mathbb{R}^m$, define*

$$\alpha = \text{sign}(x_1) \|x\|_2, \quad v = x + \alpha e_1, \quad H = I - 2 \frac{vv^T}{v^T v}.$$

Then H is orthogonal and symmetric, and

$$Hx = \begin{bmatrix} -\alpha \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

So this Householder reflector eliminates all but the first entry of x .

2.1.2 Single similarity step

Let $A^{(1)} = A$. At step j ($1 \leq j \leq n-2$), consider the subvector

$$x = A_{j+1:n,j}^{(j)}.$$

Construct a Householder reflector H_j such that $H_j x = (\pm \|x\|, 0, \dots, 0)^T$, and embed it into an $n \times n$ orthogonal matrix by

$$Q_j = \text{diag}(I_j, H_j).$$

Define

$$A^{(j+1)} = Q_j^T A^{(j)} Q_j.$$

Then all entries $A_{i,j}^{(j+1)}$ with $i \geq j+2$ vanish, and the symmetry of $A^{(j)}$ guarantees that the same entries in column j and row j are eliminated simultaneously.

2.1.3 Rank-2 update form

Let u denote the Householder vector corresponding to H_j . The similarity update on the trailing block admits the explicit rank-2 expression

$$A^{(j+1)} = A^{(j)} - uy^T - yu^T, \quad y = A^{(j)}u - \frac{1}{2}(u^T A^{(j)}u)u. \quad (\star)$$

Equation $(\star)$ is the classical symmetric Householder reduction formula.

2.1.4 Completion

Repeating the above construction for $j = 1, \dots, n-2$ yields

$$T = A^{(n-1)} = Q^T A Q, \quad Q = Q_1 Q_2 \cdots Q_{n-2},$$

where T is symmetric and tridiagonal. Since Q is orthogonal, T and A share all eigenvalues.

2.1.5 Cost

At step j , the trailing block has dimension $m = n - j$. Forming the reflector costs $O(m)$ flops, while applying the rank-2 update dominates with $\approx 4m^2$ flops. Therefore,

$$\sum_{j=1}^{n-2} 4(n-j)^2 = \frac{4}{3}n^3 + O(n^2),$$

so Householder tridiagonalization requires

$$\frac{4}{3}n^3 + O(n^2) \text{ flops.}$$

2.2 STEQR: Implicit QR/QL Iteration for Symmetric Tridiagonal Matrices

2.2.1 Problem Setting and Iteration Goal

Consider the symmetric tridiagonal eigenvalue problem

$$T = \begin{bmatrix} a_1 & b_1 & & & \\ b_1 & a_2 & b_2 & & \\ & b_2 & \ddots & \ddots & \\ & & \ddots & a_{n-1} & b_{n-1} \\ & & & b_{n-1} & a_n \end{bmatrix}$$

arising from the Householder reduction in Section 2.1. Equivalently, starting from a dense symmetric matrix A , the previous stage computes an orthogonal matrix X_0 such that

$$A_0 = X_0^T A X_0,$$

where A_0 is symmetric tridiagonal. The task of **STEQR** is then to diagonalize this tridiagonal matrix A_0 by orthogonal similarity transformations, working block by block on its active unreduced part. The goal is to compute all eigenvalues of T , and optionally its orthonormal eigenvectors, by a sequence of orthogonal similarity transformations. For exposition, this section adopts the QR viewpoint. In LAPACK, however, **STEQR/DSTEQR** may apply either implicit QR or implicit QL iteration on the current active block. In either form, the subdiagonal entries $b_i^{(k)}$ are driven toward zero; once a subdiagonal element is sufficiently small, the matrix decouples into blocks and the associated eigenvalue (or cluster) is isolated.

This section follows a single line of ideas: spectral decomposition $\rightarrow$ block iteration $\rightarrow$ QR iteration $\rightarrow$ shifted QR $\rightarrow$ tridiagonal deflation $\rightarrow$ Wilkinson shift $\rightarrow$ implicit bulge chasing $\rightarrow$ Implicit Q theorem $\rightarrow$ Francis sweep. Its purpose is to show how these ideas lead to the implicit shifted tridiagonal QR/QL iteration implemented in **STEQR**.

Notation: The symbol k denotes iteration indices, i denotes row/column indices. For a tridiagonal matrix, a_i denotes diagonal entries and b_i denotes subdiagonal entries.

2.2.2 Spectral Decomposition and Subspace Convergence

Let $A \in \mathbb{R}^{n \times n}$ be symmetric. Then it admits an orthogonal spectral decomposition

$$A = V \Lambda V^T,$$

where

$$V = [v_1, \dots, v_n], \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_n),$$

with

$$A v_j = \lambda_j v_j, \quad v_j^T v_k = \delta_{jk}, \quad |\lambda_1| \geq |\lambda_2| \geq \dots \geq |\lambda_n|.$$

For $1 \leq m \leq n-1$, define the invariant subspace

$$E_m := \text{span}\{v_1, \dots, v_m\}.$$

We first consider the action of repeated powers on a single vector. Let

$$x = \sum_{j=1}^n c_j v_j, \quad c_j = v_j^\top x.$$

Then

$$A^k x = \sum_{j=1}^n c_j \lambda_j^k v_j.$$

Thus the spectral magnitudes determine which components dominate under repeated application of A .

Let $S_m \subset \mathbb{R}^n$ be an arbitrary m -dimensional subspace. Let $X \in \mathbb{R}^{n \times m}$ be an orthonormal basis matrix whose columns span S_m . Expanding X in the eigenbasis of A , write

$$X = VC, \quad C \in \mathbb{R}^{n \times m}.$$

Repeated application of A yields

$$A^k X = V \Lambda^k C.$$

If $|\lambda_m| > |\lambda_{m+1}|$, then after normalization or orthogonalization the column space of $A^k X$ is driven toward the invariant subspace

$$E_m = \text{span}\{v_1, \dots, v_m\}.$$

Thus repeated application of A tends to amplify the dominant eigenspace components of a generic m -dimensional subspace, providing the spectral motivation for subspace iteration, block iteration, and ultimately QR iteration [6, 7].

2.2.3 Block Iteration and the QR Iteration

Starting from block subspace iteration and orthogonalization at each step, we obtain the QR form $A_{k-1} = Q_k R_k$, $A_k = R_k Q_k$. The orthogonalization is essential: without it, repeated multiplication by A would drive basis vectors toward the dominant eigendirections and lead to loss of linear independence [6].

Let $X_0 = [x_1^{(0)} \cdots x_n^{(0)}]$ be orthogonal. One block-iteration step is

$$AX_{k-1} = X_k R_k,$$

where X_k is orthogonal and R_k is upper triangular. If one were to iterate only with $A^k X_0$ and never reorthogonalize, then the columns would progressively align with the dominant eigendirections and the basis would lose linear independence. The QR step prevents this collapse by replacing the multiplied basis with an orthonormal one at every iteration while retaining the same column space. Define

$$A_k := X_k^\top A X_k.$$

Then

$$A_{k-1} = X_{k-1}^\top A X_{k-1} = X_{k-1}^\top X_k R_k = Q_k R_k, \quad Q_k := X_{k-1}^\top X_k,$$

and

$$A_k = X_k^T A X_k = X_k^T A X_{k-1} X_{k-1}^T X_k = X_k^T X_k R_k X_{k-1}^T X_k = R_k Q_k.$$

$$A_{k-1} = Q_k R_k, \quad A_k = R_k Q_k,$$

i.e., a sequence of orthogonal similarity transforms. Therefore the orthogonalized block iteration produces exactly the QR iteration [6, 8]. Since

$$A_k = Q_k^T A_{k-1} Q_k,$$

each step is an orthogonal similarity transformation and hence preserves the eigenvalues. This identity is the algebraic starting point for the shifted QR iteration used in STEQR.

2.2.4 Shifted QR Iteration

The unshifted QR iteration already preserves eigenvalues, but its convergence may be slow in practice. The standard acceleration is to introduce a shift μ_k and apply QR not to A_{k-1} itself, but to $A_{k-1} - \mu_k I$.

Introduce a shift μ_k and factor

$$A_{k-1} - \mu_k I = Q_k R_k.$$

Then update

$$A_k = \mu_k I + R_k Q_k.$$

Using the factorization above,

$$A_k = \mu_k I + R_k Q_k \tag{2.2.1}$$

$$= \mu_k I + Q_k^T (A_{k-1} - \mu_k I) Q_k \tag{2.2.2}$$

$$= Q_k^T A_{k-1} Q_k. \tag{2.2.3}$$

Therefore the shift does not alter the eigenvalue problem: each step remains an orthogonal similarity transformation. Its role is to accelerate QR iteration by choosing μ_k close to a target eigenvalue [9, 10, 6]. For symmetric tridiagonal matrices, a good shift typically speeds the decay of the trailing subdiagonal and hence accelerates deflation.

2.2.5 Symmetric Tridiagonal Case and Deflation

Let A_0 be symmetric tridiagonal. In the QR viewpoint adopted here, shifted QR is applied blockwise to active unreduced tridiagonal blocks. After each sweep, test

$$|b_i^{(k)}| \leq \tau (|a_i^{(k)}| + |a_{i+1}^{(k)}|),$$

with a machine-scaled threshold τ . If true, set $b_i^{(k)} = 0$, split the matrix into independent blocks, and continue on each block. In the QR-oriented form of the iteration, the active block is viewed from the bottom upward: the sweep is designed to drive the trailing subdiagonal toward zero, so that the bottom-right part decouples first. Thus the practical iteration repeatedly acts on the current unreduced trailing principal block, shrinks that block after deflation, and continues until only

diagonal entries remain $[2, 7]$. Recall that an upper Hessenberg matrix H satisfies $H_{ij} = 0$ for $i > j + 1$; a symmetric tridiagonal matrix is a special case.

Proposition 2.2.1 (Preservation of tridiagonal structure). *Suppose A_{k-1} is symmetric tridiagonal and*

$$A_{k-1} - \mu_k I = Q_k R_k, \quad A_k = \mu_k I + R_k Q_k.$$

Then A_k is again tridiagonal.

This standard Hessenberg/tridiagonal invariance of shifted QR is discussed in Golub–Van Loan and Parlett [6, 7].

Proof. Let $\mathcal{L}(M)$ denote the lower bandwidth of a matrix M . Since A_{k-1} is tridiagonal, $\mathcal{L}(A_{k-1}) = 1$. Subtracting $\mu_k I$ does not change bandwidth, so

$$\mathcal{L}(A_{k-1} - \mu_k I) = 1.$$

For the QR factorization of a Hessenberg or tridiagonal matrix, the orthogonal factor satisfies

$$\mathcal{L}(Q_k) \leq \mathcal{L}(A_{k-1} - \mu_k I) = 1,$$

while R_k is upper triangular, hence

$$\mathcal{L}(R_k) = 0.$$

For any $n \times n$ matrices A, B , the lower-bandwidth bound

$$\mathcal{L}(AB) \leq \mathcal{L}(A) + \mathcal{L}(B)$$

holds: from

$$(AB)_{ij} = \sum_t A_{it} B_{tj},$$

if $(AB)_{ij} \neq 0$, then for some t we have $A_{it} B_{tj} \neq 0$, so

$$i - t \leq \mathcal{L}(A), \quad t - j \leq \mathcal{L}(B),$$

and therefore

$$i - j = (i - t) + (t - j) \leq \mathcal{L}(A) + \mathcal{L}(B).$$

Therefore

$$\mathcal{L}(A_k) = \mathcal{L}(\mu_k I + R_k Q_k) = \mathcal{L}(R_k Q_k) \leq \mathcal{L}(R_k) + \mathcal{L}(Q_k) \leq 1.$$

Because A_k is symmetric and has lower bandwidth at most 1, it is tridiagonal. $\square$

2.2.6 Wilkinson Shift

In the QR-oriented form of the iteration, for the trailing 2×2 block

$$\begin{bmatrix} a & b \\ b & c \end{bmatrix}, \quad \delta := \frac{a - c}{2},$$

the Wilkinson shift is the eigenvalue closer to c , evaluated stably as

$$\mu = c - \text{sign}(\delta) \frac{b^2}{|\delta| + \sqrt{\delta^2 + b^2}},$$

with $\text{sign}(0) = 1$. This shift, commonly called the Wilkinson shift, is the standard practical choice in QR methods for symmetric tridiagonal matrices [11, 7].

2.2.7 Implicit Shift, Bulge Generation, and Bulge Chasing

Instead of explicitly forming and factoring $A_0 - \mu I$, the shift is injected implicitly through the first column

$$x := (A_0 - \mu I)e_1 = \begin{bmatrix} d_1 - \mu \\ e_1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

Because only the first two entries of x are nonzero, a single Givens rotation G_1 , acting on indices $(1, 2)$, is sufficient to map this two-vector to $(\rho, 0)^\top$. Set

$$A_0^{(1)} = G_1^\top A_0 G_1.$$

This first similarity preserves symmetry, but it no longer preserves exact tridiagonality: a new fill-in appears at $(3, 1)$ and, by symmetry, at $(1, 3)$. This isolated out-of-band entry is the *bulge*.

The local 3×3 pattern can be sketched as

$$\begin{bmatrix} \times & \times & \cdot \\ \times & \times & \times \\ \cdot & \times & \times \end{bmatrix} \xrightarrow{G_1^\top(\cdot)G_1} \begin{bmatrix} + & + & + \\ + & + & + \\ + & + & + \end{bmatrix},$$

where $\times$ denotes a known nonzero entry, $\cdot$ denotes a zero entry, $+$ denotes an entry modified by the current similarity update, and $+$ denotes the newly created bulge entry. The symbol $\times$ indicates only structural nonzero positions before the update; its numerical value may still change under the similarity transformation.

The remaining rotations are chosen only to move and eliminate this bulge while keeping the rest of the matrix tridiagonal. For $j = 2, \dots, n-1$, choose G_j on $(j, j+1)$ so that

$$A_0^{(j)} = G_j^\top A_0^{(j-1)} G_j.$$

Because G_j acts only on rows and columns j and $j+1$, it leaves the first basis vector unchanged:

$$G_j e_1 = e_1, \quad e_1^\top G_j = e_1^\top, \quad j \geq 2.$$

Hence, for $j \geq 2$,

$$e_1^\top A_0^{(j)} e_1 = e_1^\top G_j^\top A_0^{(j-1)} G_j e_1 = e_1^\top A_0^{(j-1)} e_1,$$

and likewise

$$A_0^{(j)} e_1 = G_j^T A_0^{(j-1)} e_1.$$

Thus $G_2, G_3, \dots$ do not create any new coupling with e_1 ; they only remove the current bulge and move it one position farther down. Thus $A_0^{(1)}$ has a bulge at $(3, 1)$ and, by symmetry, at $(1, 3)$. Applying G_2 gives

$$A_0^{(2)} = G_2^T A_0^{(1)} G_2,$$

which removes the bulge at $(3, 1)$ and creates the next bulge at $(4, 2)$ and $(2, 4)$. Applying G_3 gives

$$A_0^{(3)} = G_3^T A_0^{(2)} G_3,$$

which moves the bulge further to $(5, 3)$ and $(3, 5)$. In general, the bulge pair moves in the pattern

$$(3, 1) \rightarrow (4, 2) \rightarrow (5, 3) \rightarrow \dots \rightarrow (n, n-2),$$

with the symmetric counterpart in the upper triangle moving simultaneously. The first stages can be visualized as

$$\begin{aligned} A_0^{(1)} &\sim \begin{bmatrix} \times & \times & + & \cdot & \cdot \\ \times & \times & \times & \cdot & \cdot \\ + & \times & \times & \times & \cdot \\ \cdot & \cdot & \times & \times & \times \\ \cdot & \cdot & \cdot & \times & \times \end{bmatrix} \\ &\Downarrow \\ A_0^{(2)} &\sim \begin{bmatrix} \times & \times & \cdot & \cdot & \cdot \\ \times & \times & \times & + & \cdot \\ \cdot & \times & \times & \times & \cdot \\ \cdot & + & \times & \times & \times \\ \cdot & \cdot & \cdot & \times & \times \end{bmatrix} \\ &\Downarrow \\ A_0^{(3)} &\sim \begin{bmatrix} \times & \times & \cdot & \cdot & \cdot \\ \times & \times & \times & \cdot & \cdot \\ \cdot & \times & \times & \times & + \\ \cdot & \cdot & \times & \times & \times \\ \cdot & \cdot & + & \times & \times \end{bmatrix}, \end{aligned}$$

where each $+$ marks the current out-of-band fill-in and its symmetric counterpart. For an active block of order p , after $p-1$ rotations the bulge exits the block and the matrix is tridiagonal again. This is the essential reason the implicit shifted QR step is efficient: the shift is incorporated without forming a dense intermediate matrix [9, 10, 8].

2.2.8 Implicit Q Theorem and Equivalence to Explicit Shifted QR

Definition 2.2.2 (Unreduced Hessenberg matrix). *An upper Hessenberg matrix $H \in \mathbb{R}^{n \times n}$ is called unreduced if every subdiagonal entry is nonzero, that is,*

$$H_{i+1,i} \neq 0, \quad i = 1, \dots, n-1.$$

In the present tridiagonal- QR context, this means the current active block has not yet deflated: every subdiagonal entry in that block is nonzero. If some subdiagonal entry $H_{i+1,i}$ vanishes, then H is called *reduced*; the matrix splits at that position into smaller Hessenberg blocks. The unreduced assumption is needed because the Implicit Q theorem asserts uniqueness of the Hessenberg-preserving orthogonal transformation from its first column, and this uniqueness can fail once such a split occurs. For the present shifted- QR step, the matrix to which the theorem is applied is the shifted active block

$$H := A_0 - \mu I.$$

We only need the version of the theorem tailored to this situation: both the explicit QR factorization and the implicit bulge-chasing step act on the same unreduced Hessenberg matrix H , and the theorem says that once their first column agrees, the whole orthogonal factor is fixed up to signs.

Theorem 2.2.3 (Implicit Q theorem for the shifted block [8, 7]). *Let $H = A_0 - \mu I$ be unreduced upper Hessenberg, and let $\widehat{Q} = [\hat{q}_1, \dots, \hat{q}_n]$ be orthogonal such that $\widehat{Q}^T H \widehat{Q}$ is also unreduced upper Hessenberg. Then $\widehat{Q}$ is determined uniquely by its first column $\hat{q}_1$, up to columnwise signs.*

To apply this to the QR step, first consider the explicit shifted factorization

$$A_0 - \mu I = Q_e R_e,$$

where $Q_e = [q_1^{(e)}, \dots, q_n^{(e)}]$ is orthogonal and R_e is upper triangular. Since $R_e e_1 = (r_{11}, 0, \dots, 0)^T$, we have

$$(A_0 - \mu I)e_1 = Q_e R_e e_1 = r_{11} q_1^{(e)}.$$

Hence the first column of the explicit QR factor is

$$q_1^{(e)} = \pm \frac{(A_0 - \mu I)e_1}{\|(A_0 - \mu I)e_1\|_2}.$$

This is exactly the first-column vector appearing in the theorem. In the present application, the theorem's $\hat{q}_1$ is the normalized vector $(A_0 - \mu I)e_1 / \|(A_0 - \mu I)e_1\|_2$, namely the first column $q_1^{(e)}$ of the explicit QR factor Q_e , up to sign.

Because A_0 is tridiagonal, this vector is determined by the leading two entries

$$(A_0 - \mu I)e_1 = \begin{bmatrix} d_1 - \mu \\ e_1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

The implicit Francis step starts from exactly the same data: its first Givens rotation is chosen from $(d_1 - \mu, e_1)$, so the orthogonal transformation U generated by bulge chasing has the same first column as Q_e , up to sign. Equivalently, the first column supplied to the theorem is the same whether one starts from the explicit QR factorization or from the implicit bulge-chasing construction. Thus the theorem applies directly to $H = A_0 - \mu I$: since both Q_e and U reduce the same unreduced Hessenberg matrix while sharing this same first-column vector, they must agree up to signs. Therefore the implicit bulge-chasing transform is algebraically the same as the explicit shifted QR step,

$$A_0 - \mu I = Q_e R_e, \quad A_1 = \mu I + R_e Q_e = Q_e^T A_0 Q_e.$$

This is the key reason implicit shifted QR is both structure-preserving and mathematically equivalent to explicit shifted QR [9, 8].

Core Derivation of the Implicit-Shift QR Step

The central algebraic claim is that one implicit Francis sweep produces exactly the same updated matrix as one explicit shifted QR step. We now derive this claim in four steps.

Step 1: Start from the explicit shifted QR factorization. Let

$$A_0 - \mu I = QR,$$

where Q is orthogonal and R is upper triangular. The corresponding explicit shifted QR update is defined by

$$A_1 := \mu I + RQ.$$

Using $R = Q^T(A_0 - \mu I)$, we obtain

$$A_1 = \mu I + Q^T(A_0 - \mu I)Q \tag{2.2.4}$$

$$= Q^T A_0 Q. \tag{2.2.5}$$

Therefore one explicit shifted QR step is an orthogonal similarity transformation.

Step 2: Determine the first column of Q . Since A_0 is tridiagonal,

$$(A_0 - \mu I)e_1 = \begin{bmatrix} d_1 - \mu \\ e_1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \alpha q_1, \quad \alpha = \|(A_0 - \mu I)e_1\|_2.$$

Thus the shift determines the first column q_1 of the orthogonal factor.

Step 3: Construct the same first column implicitly. Choose a Givens rotation G_1 such that

$$G_1^T q_1 = e_1.$$

Since $q_1 = \pm(A_0 - \mu I)e_1 / \|(A_0 - \mu I)e_1\|_2$ has nonzero entries only in its first two positions, G_1 acts only in the $(1, 2)$ -plane. Hence

$$G_1^T \frac{(A_0 - \mu I)e_1}{\|(A_0 - \mu I)e_1\|_2} = \pm e_1,$$

so, after multiplying by G_1 , we obtain

$$G_1 e_1 = \pm \frac{(A_0 - \mu I)e_1}{\|(A_0 - \mu I)e_1\|_2} = \pm q_1.$$

Thus the first column of G_1 is already the same as the first column of the explicit QR factor, up to sign. Then

$$A_0^{(1)} = G_1^T A_0 G_1$$

creates a single fill-in at $(3, 1)$, the bulge. Continue with Givens rotations $G_2, \dots, G_{p-1}$ (for an active block of order p) that successively remove the current bulge and create the next one one position lower. After the last rotation,

$$A_1^* = U^T A_0 U, \quad U = G_1 G_2 \cdots G_{p-1},$$

and A_1^* is again tridiagonal. This follows by repeated substitution:

$$A_0^{(2)} = G_2^T A_0^{(1)} G_2 = G_2^T G_1^T A_0 G_1 G_2,$$

and continuing in the same way gives

$$A_0^{(p-1)} = G_{p-1}^T \cdots G_2^T G_1^T A_0 G_1 G_2 \cdots G_{p-1} = U^T A_0 U.$$

Moreover, for $k \geq 2$, the rotation G_k acts on indices $(k, k+1)$ and therefore leaves e_1 unchanged: $G_k e_1 = e_1$. Hence $G_2, \dots, G_{p-1}$ do not affect e_1 , so the product applied to e_1 reduces to the action of the first rotation alone. Consequently,

$$U e_1 = G_1 G_2 \cdots G_{p-1} e_1 = G_1 e_1 = \pm q_1,$$

so the full bulge-chasing transform U has the same first column as G_1 , and hence as the explicit QR factor Q , up to sign.

Step 4: Invoke the Implicit Q theorem. Because $q_1 \propto (A_0 - \mu I)e_1$, the first column of the orthogonal factor is fixed. Under the unreduced Hessenberg assumptions, the Implicit Q theorem implies that the orthogonal factor is then uniquely determined up to signs. Hence the bulge-chasing orthogonal factor U agrees with the explicit QR factor Q up to signs, and therefore

$$A_1^* = U^T A_0 U = Q^T A_0 Q = A_1.$$

Thus one implicit Francis sweep is exactly one explicit shifted QR step, but carried out in a form that preserves tridiagonal structure.

2.2.9 One Francis Sweep (Algorithmic Form)

One implicit shifted QR sweep on an active block $T_{[l:m, l:m]}$ follows:

1. Choose shift μ (typically Wilkinson from the trailing 2×2 block).
2. Form the first Givens from $(a_l - \mu, b_l)$ and create the bulge.
3. Chase the bulge using adjacent Givens rotations down to the block end.
4. Update the tridiagonal entries (a, b) and, if needed, apply the same rotations to eigenvector accumulator Z .
5. Test deflation criteria; set small subdiagonals to zero and split blocks.

This sweep is the core kernel used by **STEQR**. The complete **STEQR** iteration is obtained by repeating this kernel on the current active unreduced tridiagonal block: choose the block, compute the shift, perform one implicit Francis sweep, check whether the trailing subdiagonal has become small enough to deflate, then either split the block or continue on the remaining active part [9, 10, 2].

2.2.10 Worked Example on a Common Test Matrix

Consider the following 6×6 symmetric tridiagonal test matrix:

$$T = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 3 & 1 & 0 & 0 \\ 0 & 0 & 1 & 4 & 1 & 0 \\ 0 & 0 & 0 & 1 & 5 & 1 \\ 0 & 0 & 0 & 0 & 1 & 6 \end{bmatrix}.$$

This example instantiates the preceding theory on a concrete symmetric tridiagonal matrix. The shift is chosen from the trailing 2×2 block by the Wilkinson rule, the first Givens rotation is determined by $(T - \mu I)e_1$, and the resulting bulge-chasing process realizes one implicit shifted QR sweep. By the derivation above, this sweep is algebraically equivalent to one explicit shifted QR step while preserving tridiagonal structure. In **STEQR**, the matrix would not be stored as a full 6×6 array during this process; instead, the sweep updates only the diagonal and subdiagonal data of the current active unreduced block.

At the start, the full matrix is the active block, so the current unreduced block has order $m = 6$. One implicit shifted sweep consists of choosing a Wilkinson shift from the trailing 2×2 principal submatrix, creating a bulge from the first shifted column, chasing that bulge to the bottom, and then testing whether the trailing subdiagonal has become small enough to deflate.

Its trailing 2×2 block is

$$\begin{bmatrix} 5 & 1 \\ 1 & 6 \end{bmatrix},$$

so the Wilkinson shift is

$$\mu \approx 6.6180.$$

Following the derivation above, form the first shifted two-vector

$$x = \begin{bmatrix} d_1 - \mu \\ e_1 \end{bmatrix} = \begin{bmatrix} 1 - 6.6180 \\ 1 \end{bmatrix} \approx \begin{bmatrix} -5.6180 \\ 1 \end{bmatrix}.$$

The first rotation is a 6×6 Givens rotation acting on indices $(1, 2)$:

$$G_1 = \begin{bmatrix} c_1 & -s_1 & 0 & 0 & 0 & 0 \\ s_1 & c_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad c_1^2 + s_1^2 = 1.$$

Its active 2×2 block is therefore

$$\begin{bmatrix} c_1 & -s_1 \\ s_1 & c_1 \end{bmatrix}.$$

The parameters are chosen so that this active block satisfies

$$\begin{bmatrix} c_1 & s_1 \\ -s_1 & c_1 \end{bmatrix} \begin{bmatrix} d_1 - \mu \\ e_1 \end{bmatrix} = \begin{bmatrix} r \\ 0 \end{bmatrix},$$

This means that c_1 and s_1 are selected to annihilate the second entry of the shifted two-vector while leaving the remaining coordinates unchanged in the embedded 6×6 rotation. Equivalently,

$$r = \sqrt{(d_1 - \mu)^2 + e_1^2}, \quad c_1 = \frac{d_1 - \mu}{r}, \quad s_1 = \frac{e_1}{r},$$

with the sign convention used here. The numerical values are

$$r \approx 5.7063, \quad c_1 \approx -0.9845, \quad s_1 \approx 0.1752.$$

The first similarity

$$T^{(1)} = G_1^T T G_1$$

creates the bulge at $(3, 1)$. Numerically,

$$T^{(1)} \approx \begin{bmatrix} 0.6856 & 0.7660 & 0.1752 & 0 & 0 & 0 \\ 0.7660 & 2.3144 & -0.9845 & 0 & 0 & 0 \\ 0.1752 & -0.9845 & 3 & 1 & 0 & 0 \\ 0 & 0 & 1 & 4 & 1 & 0 \\ 0 & 0 & 0 & 1 & 5 & 1 \\ 0 & 0 & 0 & 0 & 1 & 6 \end{bmatrix},$$

so the out-of-band entry at $(3, 1)$ is directly visible. To isolate the entries modified in this first local update, the leading 3×3 window changes from

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 3 \end{bmatrix} \quad \text{to} \quad \begin{bmatrix} \mathbf{0.6856} & \mathbf{0.7660} & \mathbf{0.1752} \\ \mathbf{0.7660} & \mathbf{2.3144} & \mathbf{-0.9845} \\ \mathbf{0.1752} & \mathbf{-0.9845} & \mathbf{3} \end{bmatrix},$$

so the new fill-in appears precisely at the symmetric pair $(1, 3)$ and $(3, 1)$.

To make the chase itself visible, apply the next Givens rotation G_2 on indices $(2, 3)$. This removes the bulge at $(3, 1)$ and creates the next bulge at $(4, 2)$. Numerically,

$$T^{(2)} \approx \begin{bmatrix} 0.6856 & 0.7858 & 0 & 0 & 0 & 0 \\ 0.7858 & 1.9204 & -0.7376 & 0.2230 & 0 & 0 \\ 0 & -0.7376 & 3.3939 & 0.9748 & 0 & 0 \\ 0 & 0.2230 & 0.9748 & 4 & 1 & 0 \\ 0 & 0 & 0 & 1 & 5 & 1 \\ 0 & 0 & 0 & 0 & 1 & 6 \end{bmatrix},$$

so the bulge has moved from $(3, 1)$ to $(4, 2)$. The later rotations are constructed in exactly the same way: at each stage, G_k is built from the two entries containing the current bulge, and it is chosen so that the lower entry is annihilated. This removes the present bulge and creates the next one one position farther down. Equivalently, the affected leading 4×4 window changes from

$$\begin{bmatrix} 0.6856 & 0.7660 & 0.1752 & 0 \\ 0.7660 & 2.3144 & -0.9845 & 0 \\ 0.1752 & -0.9845 & 3 & 1 \\ 0 & 0 & 1 & 4 \end{bmatrix} \quad \text{to} \quad \begin{bmatrix} 0.6856 & \mathbf{0.7858} & \mathbf{0} & \mathbf{0} \\ \mathbf{0.7858} & \mathbf{1.9204} & \mathbf{-0.7376} & \mathbf{0.2230} \\ \mathbf{0} & \mathbf{-0.7376} & \mathbf{3.3939} & \mathbf{0.9748} \\ 0 & \mathbf{0.2230} & \mathbf{0.9748} & 4 \end{bmatrix},$$

so the old bulge at $(3, 1)$ is eliminated, while the new symmetric fill-in appears at $(2, 4)$ and $(4, 2)$.

The same mechanism repeats: G_3 moves the bulge to $(5, 3)$, G_4 moves it to $(6, 4)$, and G_5 removes the last bulge so that it exits the active block. Hence the numerical sequence itself already shows the bulge-chasing process without needing a separate schematic. This is the essential idea of implicit shifted QR: the shift is encoded through the first local Givens rotation, and the remaining orthogonal transformation is realized by bulge chasing, so the method produces the shifted QR update without ever forming a dense factorization of $T - \mu I$.

The numerical data for the first sweep are summarized in Table 2.1.

Step	indices	input vector	(c_k, s_k)	bulge after step
G_1	$(1, 2)$	$(-5.6180, 1)$	$(-0.9845, 0.1752)$	$(3, 1)$
G_2	$(2, 3)$	$(0.7660, 0.1752)$	$(0.9748, 0.2230)$	$(4, 2)$
G_3	$(3, 4)$	$(-0.7376, 0.2230)$	$(-0.9572, 0.2894)$	$(5, 3)$
G_4	$(4, 5)$	$(0.6436, 0.2894)$	$(0.9120, 0.4101)$	$(6, 4)$
G_5	$(5, 6)$	$(-0.4442, 0.4101)$	$(-0.7347, 0.6783)$	exits

Table 2.1: Numerical data for one implicit Francis sweep on the 6×6 test matrix.

Thus the bulge path is

$$(3, 1) \rightarrow (4, 2) \rightarrow (5, 3) \rightarrow (6, 4) \rightarrow \text{exit}.$$

After the final rotation, tridiagonality is restored:

$$T^+ = (G_1 G_2 G_3 G_4 G_5)^T T (G_1 G_2 G_3 G_4 G_5).$$

The first sweep changes the tridiagonal data from

$$d = [1, 2, 3, 4, 5, 6], \quad e = [1, 1, 1, 1, 1]$$

to

$$d^+ \approx [0.6856, 1.9204, 2.9046, 3.8592, 4.8912, 6.7390],$$

$$e^+ \approx [0.7858, 0.7705, 0.7057, 0.6046, -0.1116].$$

In particular, the trailing subdiagonal is reduced from

$$|e_5| = 1 \quad \text{to} \quad |e_5^+| \approx 1.1164 \times 10^{-1},$$

which is the quantitative signal that the sweep is pushing the active block toward deflation.

For this test matrix, repeated Wilkinson-shift sweeps reduce the subdiagonal entries rapidly. A short summary is

sweep	0	1	2	3
$ e_5 $	1	1.1164×10^{-1}	3.4294×10^{-5}	5.0295×10^{-16}

so the trailing entry deflates after only a few sweeps. In the practical STEQR loop, once $|e_5|$ falls below the deflation threshold, the 6×6 active block splits into a 5×5 leading block and a 1×1 trailing block, and the process continues on the remaining active block. This example therefore shows the complete logic of STEQR in miniature: choose a shift, create a bulge, chase it, restore tridiagonality, test the trailing subdiagonal, and deflate.

If eigenvectors are requested, the same rotations are accumulated sweep by sweep. After one sweep on the active block, the local eigenvector accumulator is

$$Z^{(1)} = G_1 G_2 G_3 G_4 G_5 =: U,$$

so that

$$T^+ = (Z^{(1)})^T T Z^{(1)}.$$

Here $Z^{(1)}$ is exactly the accumulated orthogonal transformation produced by the implicit sweep. In the notation of the derivation above, this product is the bulge-chasing orthogonal factor U . By the Implicit Q theorem, U agrees, up to signs, with the orthogonal factor Q in the explicit shifted QR factorization

$$T - \mu I = QR.$$

After repeated sweeps and deflations, the full iteration produces

$$T^{(k)} \rightarrow \text{diag}(\lambda_1, \dots, \lambda_6), \quad Z^{(k)} \rightarrow [z_1, \dots, z_6],$$

where the columns z_i are the orthonormal eigenvectors of the tridiagonal matrix. Thus the practical output of STEQR is a diagonal matrix of converged eigenvalues together with, when requested, the accumulated eigenvector matrix.

With shifts, eigenvalues may converge and deflate in a non-monotone order, so they are not generally obtained in sorted order during the iteration. For this matrix, the eigenvalues are reported below in ascending order:

$$\lambda \approx [0.2538, 1.7895, 2.9649, 4.0351, 5.2105, 6.7462].$$

To illustrate the eigenvector output without listing the entire 6×6 orthogonal matrix, one may show one representative rounded column. For the smallest eigenvalue $\lambda_1 \approx 0.2538$, the associated normalized eigenvector is approximately

$$z_1 \approx \begin{bmatrix} 0.7800 \\ -0.5800 \\ 0.2400 \\ -0.0700 \\ 0.0100 \\ 0 \end{bmatrix}.$$

Hence the example exhibits both outputs of **STEQR**: the converged eigenvalues and, when eigenvectors are requested, columns of the accumulated orthogonal matrix.

2.2.11 Practical Remarks

One Francis sweep should be distinguished from the full **STEQR** iteration. A single sweep applies one implicit shifted QR step to the current active unreduced block. The complete solver then repeats the same sweep–deflation loop until all active blocks have become diagonal.

If only eigenvalues are required, the iteration updates only the tridiagonal data (a, b) . If eigenvectors are also required, the same Givens rotations are accumulated into an orthogonal matrix Z , so that the final dense eigenvectors are obtained as $Q_{\text{trd}}Z$ after the initial tridiagonalization. This accumulation preserves orthogonality but also makes the full-eigenvector path more sequential.

For a symmetric tridiagonal matrix, one implicit-shift sweep costs $O(n)$, so the eigenvalue-only path costs $O(n^2)$ over all sweeps. The method is backward stable provided the implementation uses robust Givens constructions, a cancellation-safe Wilkinson formula, and machine-scaled deflation tests [2, 7].

Implicit QR/QL on a tridiagonal matrix is therefore robust and memory-light, but less attractive for large full-eigenvector problems because of its long sequential rotation chains. This limitation motivates divide-and-conquer (**STEDC**), developed in Section 2.3.

2.3 STEDC: Divide-and-Conquer Method for Tridiagonal Eigenproblems

2.3.1 Motivation and Roadmap

Classical QR/QL iteration, as discussed in Section 2.2, is robust and mathematically elegant, but on the full eigenvector path it is also sequential and expensive. A standard back-of-the-envelope estimate is about $6n^3$ flops: each implicit-shift sweep applies about n Givens rotations to an $n \times n$ eigenvector matrix, and about n sweeps

are needed in total [2, 7]. This structure offers limited opportunity for concurrency because each sweep depends on the previous one.

The divide-and-conquer (D&C) strategy addresses the same tridiagonal eigenproblem from a different angle. Instead of evolving the matrix through many local QR sweeps, it first splits the tridiagonal matrix into smaller independent subproblems and then merges their eigensystems through a rank-one correction. In this form, the dominant work is reorganized into larger matrix operations, which is why the method becomes attractive when both eigenvalues and eigenvectors are computed [3, 4].

This section follows a single line of ideas: split the tridiagonal matrix, diagonalize the subproblems, reduce the merge problem to a diagonal-plus-rank-one form, solve the secular equation, reconstruct the eigenvectors, and finally explain why deflation and recursion make the method effective in practice. Its purpose is not to reproduce every implementation detail of DSTEDC, but to explain the mathematical structure that later reappears in the LAPACK implementation.

Notation: For a symmetric tridiagonal matrix, a_i denotes diagonal entries and b_i subdiagonal entries. Standard basis vectors are still written e_i . The scalar ρ denotes the coupling removed at the splitting point.

2.3.2 Divide Step: Splitting into Compensated Blocks

Consider a real symmetric tridiagonal matrix

$$T = \begin{bmatrix} a_1 & b_1 & & & \\ b_1 & a_2 & b_2 & & \\ & b_2 & \ddots & \ddots & \\ & & \ddots & a_{n-1} & b_{n-1} \\ & & & b_{n-1} & a_n \end{bmatrix} \in \mathbb{R}^{n \times n}.$$

Choose a splitting index $m \approx n/2$ and isolate the coupling $\rho := b_m$. If one first separates the matrix into its two principal blocks, the raw decomposition is

$$T = \text{blkdiag}(\hat{T}_1, \hat{T}_2) + \rho(e_m e_{m+1}^T + e_{m+1} e_m^T),$$

where $\hat{T}_1 = T(1:m, 1:m)$ and $\hat{T}_2 = T(m+1:n, m+1:n)$. This already decouples the two halves structurally, but the coupling term is rank two rather than rank one.

The key algebraic trick is therefore to absorb a diagonal correction into each block. Define the compensated tridiagonal matrices

$$T_1 = \hat{T}_1 - \rho e_m e_m^T, \quad T_2 = \hat{T}_2 - \rho e_1 e_1^T,$$

where $e_m \in \mathbb{R}^m$ and $e_1 \in \mathbb{R}^{n-m}$ are local standard basis vectors. With

$$u = \begin{bmatrix} e_m \\ e_1 \end{bmatrix},$$

the identity

$$uu^T = \begin{bmatrix} e_m e_m^T & e_m e_1^T \\ e_1 e_m^T & e_1 e_1^T \end{bmatrix}$$

shows the following standard reformulation.

Proposition 2.3.1 (Compensated Split and Rank-One Reduction). *Let T be a symmetric tridiagonal matrix and let $\rho = b_m$ be the subdiagonal coupling at the chosen splitting index m . With the compensated blocks T_1, T_2 defined above and $u = \begin{bmatrix} e_m \\ e_1 \end{bmatrix}$, one has*

$$T = \text{blkdiag}(T_1, T_2) + \rho uu^\top. \quad (2.3.1)$$

If, in addition, $T_1 = Q_1 \Lambda_1 Q_1^\top$ and $T_2 = Q_2 \Lambda_2 Q_2^\top$, then with $Q_0 = \text{blkdiag}(Q_1, Q_2)$ and $v = Q_0^\top u$, the transformed problem satisfies

$$Q_0^\top T Q_0 = (\Lambda_1 \oplus \Lambda_2) + \rho vv^\top.$$

This is the standard Cuppen reformulation, later stabilized and refined by Gu–Eisenstat [3, 4]. The split is useful not merely because it produces two smaller matrices, but because it converts their coupling into a form that can later be analyzed through a scalar equation.

Proof. By construction,

$$\text{blkdiag}(T_1, T_2) = \text{blkdiag}(\widehat{T}_1, \widehat{T}_2) - \rho \text{blkdiag}(e_m e_m^\top, e_1 e_1^\top).$$

Since

$$uu^\top = \begin{bmatrix} e_m e_m^\top & e_m e_1^\top \\ e_1 e_m^\top & e_1 e_1^\top \end{bmatrix},$$

adding $\rho uu^\top$ restores the two interface diagonal terms and introduces the off-diagonal coupling

$$\rho \begin{bmatrix} 0 & e_m e_1^\top \\ e_1 e_m^\top & 0 \end{bmatrix},$$

which is exactly the coupling between the two raw principal blocks at the split location. Hence

$$\text{blkdiag}(T_1, T_2) + \rho uu^\top = \text{blkdiag}(\widehat{T}_1, \widehat{T}_2) + \rho (e_m e_{m+1}^\top + e_{m+1} e_m^\top) = T.$$

For the transformed form, use

$$\text{blkdiag}(T_1, T_2) = Q_0 (\Lambda_1 \oplus \Lambda_2) Q_0^\top$$

with $Q_0 = \text{blkdiag}(Q_1, Q_2)$, and multiply (2.3.1) on the left by $Q_0^\top$ and on the right by Q_0 . This gives

$$Q_0^\top T Q_0 = (\Lambda_1 \oplus \Lambda_2) + \rho (Q_0^\top u)(Q_0^\top u)^\top,$$

that is,

$$Q_0^\top T Q_0 = (\Lambda_1 \oplus \Lambda_2) + \rho vv^\top. \square$$

$$\left[\begin{array}{c|c} \hat{T}_1 & \rho e_m e_1^\top \\ \hline \rho e_1 e_m^\top & \hat{T}_2 \end{array} \right] \Rightarrow \text{blkdiag}(T_1, T_2) + \rho u u^\top, \quad u = \begin{bmatrix} e_m \\ e_1 \end{bmatrix}.$$

Figure 2.1: Schematic split at the interface between rows and columns m and $m + 1$. The raw principal blocks are first separated and then compensated so that the remaining coupling becomes a rank-one correction.

Running Example. To keep the discussion concrete, reuse the 6×6 tridiagonal test matrix from Section 2.2:

$$T = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 3 & 1 & 0 & 0 \\ 0 & 0 & 1 & 4 & 1 & 0 \\ 0 & 0 & 0 & 1 & 5 & 1 \\ 0 & 0 & 0 & 0 & 1 & 6 \end{bmatrix}, \quad m = 3, \quad \rho = 1.$$

The raw principal blocks are

$$\hat{T}_1 = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 3 \end{bmatrix}, \quad \hat{T}_2 = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 5 & 1 \\ 0 & 1 & 6 \end{bmatrix}.$$

After compensation,

$$T_1 = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}, \quad T_2 = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 5 & 1 \\ 0 & 1 & 6 \end{bmatrix},$$

and

$$u = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}.$$

The example is small enough that the full algebra can be written down explicitly, but it still shows the same split-and-merge mechanism used at larger sizes.

2.3.3 Subproblem Diagonalization and Recursive Structure

Once the compensated split has been formed, the two subproblems are diagonalized independently:

$$T_1 = Q_1 \Lambda_1 Q_1^\top, \quad T_2 = Q_2 \Lambda_2 Q_2^\top, \quad Q_i^\top Q_i = I. \quad (2.3.2)$$

If a subproblem is still too large, the same splitting idea is applied recursively. The divide stage therefore produces a binary tree of smaller tridiagonal eigenproblems, terminated only when the leaf blocks are small enough to be solved directly.

At this stage the subproblems are completely independent. This independence is the first source of parallelism in the D&C method, but it is also important mathematically: once T_1 and T_2 have been diagonalized, all remaining global information about the original matrix must enter through the rank-one coupling in (2.3.1). The merge step is therefore not a generic combination of two eigensystems; it is a highly structured correction to two already diagonal pieces.

Running Example. For the 6×6 example, the eigendecompositions of the two 3×3 blocks yield

$$\Lambda_1 = \text{diag}(0.1981, 1.5550, 3.2470), \quad \Lambda_2 = \text{diag}(2.5395, 4.7609, 6.6996).$$

Only the boundary vectors $Q_1^T e_3$ and $Q_2^T e_1$ are needed for the later merge. Numerically,

$$Q_1^T e_3 \approx \begin{bmatrix} 0.3280 \\ 0.7370 \\ 0.5910 \end{bmatrix}, \quad Q_2^T e_1 \approx \begin{bmatrix} 0.9018 \\ 0.4042 \\ 0.1531 \end{bmatrix}.$$

These six numbers carry the entire influence of the original coupling into the merge stage.

2.3.4 Merge Step: Diagonal-Plus-Rank-One Form

Assemble

$$Q_0 := \text{blkdiag}(Q_1, Q_2), \quad D := \Lambda_1 \oplus \Lambda_2.$$

Applying the block-diagonal orthogonal matrix Q_0 to Equation (2.3.1) gives

$$Q_0^T T Q_0 = Q_0^T \text{blkdiag}(T_1, T_2) Q_0 + \rho (Q_0^T u)(Q_0^T u)^T \quad (2.3.3)$$

$$= \text{blkdiag}(\Lambda_1, \Lambda_2) + \rho v v^T, \quad (2.3.4)$$

where

$$v := Q_0^T u = \begin{bmatrix} Q_1^T e_m \\ Q_2^T e_1 \end{bmatrix}. \quad (2.3.5)$$

Hence

$$Q_0^T T Q_0 = D + \rho v v^T. \quad (2.3.6)$$

This is the key structural reduction of the method: after the split and local diagonalizations, the global merge problem is no longer tridiagonal, but it is only a rank-one modification of a diagonal matrix. The split by itself is not yet enough; what matters is that the remaining interaction has the special form $D + \rho v v^T$, for which the eigenvalues can be recovered from a one-dimensional secular equation.

Equation (2.3.5) also shows why only boundary information from the subproblems is needed. The vectors $Q_1^T e_m$ and $Q_2^T e_1$ are just the last row of Q_1 and the first row of Q_2 , written as column vectors. In other words, after diagonalizing the subproblems, the merge does not need the full matrices T_1 and T_2 again; it only needs their eigenvalues and the way their eigenvectors touch the split interface.

For the secular analysis it is convenient to sort the diagonal entries of D . If P is a permutation matrix such that $P^T D P$ is diagonal with ascending entries, then

$$P^T(D + \rho v v^T)P = (P^T D P) + \rho (P^T v)(P^T v)^T.$$

Hence reordering does not change the structure of the merge problem; it merely relabels the poles and applies the same permutation to the coupling vector.

Running Example. For the 6×6 example,

$$D = \text{diag}(0.1981, 1.5550, 3.2470, 2.5395, 4.7609, 6.6996),$$

and the transformed update vector is

$$v \approx \begin{bmatrix} 0.3280 \\ 0.7370 \\ 0.5910 \\ 0.9018 \\ 0.4042 \\ 0.1531 \end{bmatrix}.$$

At this point, the original tridiagonal problem has been reduced completely to the eigenproblem for $D + \rho v v^T$.

2.3.5 Secular Equation

Consider the eigenproblem

$$(D + \rho v v^T)y = \lambda y.$$

Assume first that $\lambda \neq d_i$ for every diagonal entry d_i of D . Rearranging gives

$$(\lambda I - D)y = \rho v(v^T y).$$

The right-hand side is a scalar multiple of v , so any nonzero eigenvector y must be parallel to $(\lambda I - D)^{-1}v$. Writing $\alpha := \rho(v^T y)$, one obtains

$$y = \alpha(\lambda I - D)^{-1}v.$$

Multiplying by v^T and dividing by the nonzero scalar $v^T y$ yields

$$1 = \rho v^T (\lambda I - D)^{-1} v = \rho \sum_{i=1}^n \frac{v_i^2}{\lambda - d_i}. \quad (2.3.7)$$

Equivalently, the eigenvalues are the roots of the secular equation

$$f(\lambda) := 1 - \rho \sum_{i=1}^n \frac{v_i^2}{\lambda - d_i} = 0. \quad (2.3.8)$$

Thus the $n \times n$ eigenproblem for $D + \rho v v^T$ is reduced to a scalar root-finding problem for a rational function. The next proposition summarizes the standard secular characterization used in divide-and-conquer.

To see the geometric structure of the roots, reorder the diagonal entries if necessary and write

$$\tilde{D} := P^T D P = \text{diag}(\delta_1, \dots, \delta_n), \quad \delta_1 < \delta_2 < \dots < \delta_n.$$

Let $\tilde{v} := P^T v$. Then the secular equation becomes

$$1 - \rho \sum_{i=1}^n \frac{\tilde{v}_i^2}{\lambda - \delta_i} = 0.$$

Proposition 2.3.2 (Secular Equation and Interlacing). *Let $\tilde{D} = \text{diag}(\delta_1, \dots, \delta_n)$ with $\delta_1 < \delta_2 < \dots < \delta_n$, and consider the active nondeflated rank-one modification $\tilde{D} + \rho \tilde{v} \tilde{v}^T$ with $\rho > 0$. Then its eigenvalues are exactly the roots of*

$$1 - \rho \sum_{i=1}^n \frac{\tilde{v}_i^2}{\lambda - \delta_i} = 0.$$

Moreover, these roots strictly interlace the poles:

$$\delta_1 < \lambda_1 < \delta_2 < \lambda_2 < \dots < \delta_n < \lambda_n.$$

This characterization and the associated interlacing behavior are standard for rank-one modifications of diagonal matrices in the divide-and-conquer literature [7, 4]. For the active nondeflated problem with $\rho > 0$, the proof is a direct consequence of the monotonicity of the secular function between consecutive poles. Indeed,

$$f'(\lambda) = \rho \sum_{i=1}^n \frac{\tilde{v}_i^2}{(\lambda - \delta_i)^2} > 0$$

away from the poles, so f is strictly increasing on each interval between two consecutive poles. Moreover,

$$f(\lambda) \rightarrow -\infty \quad \text{as } \lambda \downarrow \delta_i, \quad f(\lambda) \rightarrow +\infty \quad \text{as } \lambda \uparrow \delta_{i+1}.$$

Therefore there is exactly one root in each interval (δ_i, δ_{i+1}) , and one additional root in (δ_n, ∞) . In particular,

$$\delta_1 < \lambda_1 < \delta_2 < \lambda_2 < \dots < \delta_n < \lambda_n.$$

This interlacing property is one of the main reasons the secular equation is numerically manageable.

Running Example. For the 6×6 example, it is convenient to sort the diagonal entries and permute the components of v consistently. If P denotes the associated permutation matrix, then $P^T D P$ and $P^T v$ are

$$\tilde{D} = \text{diag}(0.1981, 1.5550, 2.5395, 3.2470, 4.7609, 6.6996),$$

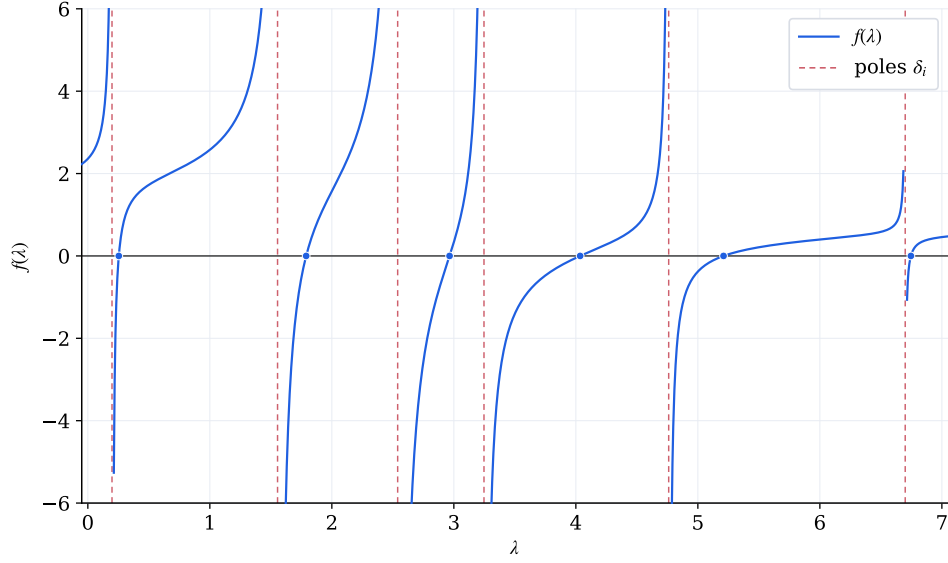


Figure 2.2: Secular function $f(\lambda)$ for the 6×6 running example. The dashed lines mark the poles δ_i , while the blue intersections with the horizontal axis indicate the roots λ_i .

$$\tilde{v} \approx \begin{bmatrix} 0.3280 \\ 0.7370 \\ 0.9018 \\ 0.5910 \\ 0.4042 \\ 0.1531 \end{bmatrix}, \quad \rho = 1.$$

The secular function is therefore approximately

$$f(\lambda) \approx 1 - \frac{0.1076}{\lambda - 0.1981} - \frac{0.5431}{\lambda - 1.5550} - \frac{0.8132}{\lambda - 2.5395} - \frac{0.3493}{\lambda - 3.2470} - \frac{0.1634}{\lambda - 4.7609} - \frac{0.0234}{\lambda - 6.6996}.$$

Solving $f(\lambda) = 0$ on the interlacing intervals gives

$$\lambda \approx [0.2538, 1.7895, 2.9649, 4.0351, 5.2105, 6.7462].$$

Figure 2.2 does not replace the argument above, but it makes the interlacing mechanism visible for the running example. The dashed vertical lines mark the poles δ_i , and the graph is monotone between consecutive poles and crosses the horizontal axis exactly once in each admissible interval. What matters here is not the visual steepness itself, but the fact that $f'(\lambda) > 0$ between consecutive poles, which guarantees monotonicity and hence uniqueness of the root in each interlacing interval.

2.3.6 Eigenvector Reconstruction

Once a root λ_j of the secular equation has been found, the corresponding eigenvector in the diagonal basis is obtained from

$$\tilde{y}_j \propto (\lambda_j I - \tilde{D})^{-1} \tilde{v}.$$

If a permutation P was introduced to order the poles, then the eigenvector in the original merge basis is $y_j = P\tilde{y}_j$. After normalization, the eigenvectors of the original tridiagonal matrix follow by transforming back with Q_0 :

$$z_j = Q_0 P \tilde{y}_j, \quad T z_j = \lambda_j z_j. \quad (2.3.9)$$

Collecting all reconstructed vectors into $\tilde{Y} = [\tilde{y}_1, \dots, \tilde{y}_n]$ and $Z = [z_1, \dots, z_n]$, one obtains the full spectral decomposition

$$T = Z \Lambda Z^T, \quad Z = Q_0 P \tilde{Y}.$$

This reconstruction formula explains why the D&C method is especially attractive when eigenvectors are needed. The expensive work is not a long chain of local rotations, as in the QR route, but a structured sequence of matrix-vector and matrix-matrix operations arising from the recursive merges. At each merge level, the new eigenvectors are assembled from previously computed local eigenvectors and the small amount of coupling data encoded in v .

Running Example. For the smallest eigenvalue $\lambda_1 \approx 0.2538$, the vector $(\lambda_1 I - \tilde{D})^{-1} \tilde{v}$ gives the corresponding eigenvector in the ordered diagonal basis. After normalization, undoing the permutation, and applying the back transformation, one obtains a representative eigenvector of the original matrix:

$$z_1 \approx \begin{bmatrix} 0.7770 \\ -0.5798 \\ 0.2354 \\ -0.0667 \\ 0.0146 \\ -0.0025 \end{bmatrix}.$$

As usual, the sign is not unique: both z_1 and $-z_1$ represent the same eigenspace.

2.3.7 Deflation and Reordering

In practical D&C implementations, the secular equation is usually not solved for all n indices simultaneously. Instead, one first looks for deflation, that is, situations in which part of the problem has already decoupled numerically. Two mechanisms are especially important:

- If a component v_i is zero, or numerically negligible, then the corresponding pole d_i is already an eigenvalue of $D + \rho v v^T$, so that index can be removed from the active secular solve.
- If two diagonal entries d_i and d_j are very close, a local orthogonal rotation can be used to combine them so that one component of v vanishes. This idea lies behind the Gu–Eisenstat stabilization strategy [4].

Deflation and reordering are therefore not secondary implementation tricks. They are part of the mathematical mechanism that turns the naive rank-one merge into a stable and efficient algorithm. Reordering places the poles in the monotone order required by the interlacing analysis, while deflation shrinks the active secular problem and improves robustness in clustered spectra.

For the small 6×6 running example above, no substantial deflation occurs, so the secular equation remains six-dimensional. In larger problems, however, deflation can significantly reduce the work and is one reason why the practical cost is often better than the worst-case recurrence suggests.

2.3.8 Complexity and Parallel Structure

From the complexity point of view, the important case is again the computation of both eigenvalues and eigenvectors. In a simplified asymptotic model for that path, ignoring deflation and lower-order leaf costs, the dominant merge work satisfies

$$C(n) = n^3 + 2C(n/2).$$

The term n^3 reflects the cost of combining eigenvector information at a merge node, while the two recursive terms correspond to the two half-size subproblems. Expanding the recursion tree, at level j there are 2^j subproblems of size $n/2^j$, so the total merge work on that level is

$$2^j \left(\frac{n}{2^j} \right)^3 = \frac{n^3}{4^j}.$$

Summing over all levels gives

$$\sum_{j \geq 0} \frac{n^3}{4^j} < \frac{4}{3} n^3,$$

so in this model

$$C(n) \sim \frac{4}{3} n^3.$$

This should be compared with the rough $6n^3$ estimate for the QR route from Section 2.2. The constants are only approximate, and practical performance depends on deflation and implementation details, but the structural contrast is already clear.

The same recursion also exposes useful parallelism. The two child problems can be solved independently, and the merge stage is dominated by larger dense operations rather than long chains of dependent Givens rotations. This is the structural reason why divide-and-conquer later aligns better with BLAS-3-oriented implementations in LAPACK. The running 6×6 example is far too small to display this asymptotic behavior, but it exhibits the same split/merge pattern on which the large-scale cost analysis rests.

2.3.9 Summary

The divide-and-conquer method solves the same tridiagonal eigenproblem as the QR route, but it proceeds through a different mathematical decomposition:

1. split the tridiagonal matrix into two compensated subproblems,
2. diagonalize the subproblems independently and recursively if needed,
3. transform the merge problem into $D + \rho vv^T$,

4. solve the secular equation and reconstruct eigenvectors,
5. reduce the active problem further through deflation and reordering.

In contrast with **STEQR**, which advances by local implicit-shift sweeps and repeated deflation of active tridiagonal blocks, **STEDC** proceeds by recursive splitting and structured rank-one merging. That contrast explains why the QR route is more sequential, while divide-and-conquer is more global and better aligned with larger matrix operations. The next section compares the two procedures directly, and Chapter 3 shows how the same distinction reappears in the LAPACK drivers **DSYEV** and **DSYEVD**.

2.4 Comparison of Algorithmic Procedures

Let

$$T \in \mathbb{R}^{n \times n}, \quad T = T^T$$

be tridiagonal. Sections 2.2 and 2.3 have presented two different ways to compute its eigendecomposition. The purpose of this section is not to introduce new mathematics, but to summarize the main differences between the QR-based route (**STEQR**) and the divide-and-conquer route (**STEDC**) before turning to the LAPACK implementation in Chapter 3.

2.4.1 Conceptual Comparison

In exact arithmetic, both methods aim at the same final object,

$$T = Z \Lambda Z^T,$$

so the mathematical target is identical. The difference lies in how the computation moves from the input tridiagonal matrix to that final spectral factorization.

STEQR proceeds by local implicit iteration. Each active unreduced block is updated through shifted QR/QL sweeps, with Givens rotations chasing a bulge through the tridiagonal structure until small subdiagonal entries deflate. Eigenvectors, when requested, are obtained by accumulating those rotations.

STEDC, by contrast, proceeds by global split-and-merge. The matrix is partitioned into smaller subproblems, each subproblem is solved separately, and the solutions are recombined through a rank-one correction and the secular equation. Eigenvectors are then reconstructed from the merged subproblem data.

The main contrast is summarized in Table 2.2.

Table 2.2: Conceptual comparison between **STEQR** and **STEDC**.

Aspect	STEQR	STEDC
Primary state	active tridiagonal block	$D + \rho v v^T$
Main nonlinear step	implicit shift + bulge chase	secular equation solve
Orthogonal map	$\prod_k G_k$	$Q_0 P Y$
Stopping signal	$ b_i $ deflation	secular residual/interlacing

The key distinction is that **STEQR** evolves the solution through many fine-grained local updates, whereas **STEDC** reorganizes the problem into subproblems and structured merge operations.

2.4.2 Complexity and Numerical Stability

From the complexity point of view, the two methods differ most clearly when eigenvectors are required. For eigenvalues only, STEQR on a tridiagonal matrix is typically $O(n^2)$, since each implicit sweep costs $O(n)$ and many such sweeps may be needed. On the full eigenvector path, the repeated accumulation of plane rotations leads to cubic total work.

For STEDC, the divide-and-conquer model gives the recurrence

$$C(n) = n^3 + 2C(n/2),$$

so the leading-order cost is

$$C(n) \sim \frac{4}{3}n^3,$$

ignoring reductions from deflation. This estimate explains why divide-and-conquer becomes attractive for large full-eigenvector problems: its dominant work is organized around larger matrix operations and recursive subproblems, rather than long chains of sequential rotations.

The two methods also concentrate numerical difficulty in different places. STEQR is generally very robust, but its sensitivity is tied to long rotation chains, shift selection, and the deflation tests that decide when a subdiagonal entry is negligible. STEDC also preserves symmetry and orthogonality well, but its most delicate step is the secular solve, particularly when poles are tightly clustered and deflation or reordering choices become important.

2.4.3 Conclusion

STEQR and STEDC therefore solve the same tridiagonal eigenproblem but follow markedly different computational routes. The QR path is more local, sequential, and memory-light; the divide-and-conquer path is more global, recursive, and better aligned with large-scale eigenvector computations. This contrast forms the bridge to Chapter 3, where the same structural difference reappears in the LAPACK drivers DSYEV and DSYEVD.

CHAPTER 3

Implementation-Level Comparison of DSYEV and DSYEVD

This chapter bridges the mathematical algorithms of Chapter 2 and the performance results of Chapter 4. Its purpose is not to reproduce LAPACK documentation, but to explain the implementation choices that matter for understanding the practical difference between DSYEV and DSYEVD.

The focus is therefore on implementation structure, data flow, and the stages where the two drivers genuinely diverge. Section 3.1 gives an overview of the two routines and the common three-stage pipeline. Section 3.2 then examines the key implementation blocks, with emphasis on the shared reduction DSYTRD, the QR-based tridiagonal solver DSTEQR, and the divide-and-conquer solver DSTEDC. Section 3.3 summarizes the main implementation implications and explains why these differences matter for the performance study in Chapter 4.

3.1 Overview of the Driver Structure

This section gives a compact overview of DSYEV and DSYEVD before the chapter turns to their main implementation differences. The aim is not to document every internal subroutine, but to identify the small number of design choices that are most relevant for the performance discussion in Chapter 4.

3.1.1 Interface-Level View

At the user level, the two LAPACK routines answer the same basic question: given a real symmetric dense matrix A , compute its eigenvalues and, optionally, its eigenvectors. In both cases the mathematical output is

$$AU = U\Lambda,$$

with the eigenvalues returned in ascending order and the eigenvectors orthonormal.

At the interface level, the most important points are the following:

- JOBZ determines whether only eigenvalues or both eigenvalues and eigenvectors are returned.
- UPLO specifies whether the upper or lower triangular part of the symmetric matrix is supplied.
- DSYEVD solves the same problem as DSYEV, but uses additional workspace in order to employ the divide-and-conquer algorithm.

The two routines are summarized in Table 3.1.

For the present chapter, the external interfaces are not the main source of interest, since they are very similar in both cases. The significant distinction appears after the driver begins its internal work: DSYEV follows a QR-based tridiagonal path, whereas DSYEVD switches to divide-and-conquer.

Table 3.1: Interface-level summary of DSYEV and DSYEVD.

Routine	Main idea	Relevant interface features
DSYEV	Classical QR-based symmetric eigensolver for dense matrices.	Same input/output goal as DSYEVD; smaller workspace; most relevant parameters here are JOBZ, UPLO, and WORK/LWORK.
DSYEVD	Divide-and-conquer version of the same dense symmetric eigensolver.	Interface is almost the same as DSYEV, but it needs larger workspace, including IWORK/LIWORK; this extra storage usually pays off when all eigenvectors are needed.

3.1.2 Common Driver Pipeline

Both drivers follow the same high-level three-stage pipeline; they differ only in the routines used after the shared reduction stage:

1. **Shared front end:** DSYTRD reduces the dense symmetric matrix A to tridiagonal form.
2. **Main point of divergence:** DSYEV uses DSTEQR, while DSYEVD uses DSTEDC.
3. **Eigenvector recovery:** when $JOBZ='V'$, DSYEV forms Q explicitly through DORGTR, whereas DSYEVD applies the stored reflectors later through DORMTR.

The important message is simple: the shared reduction stage does not explain most of the performance gap. The real difference appears in the tridiagonal eigensolver and in how the tridiagonal eigenvectors are mapped back to the dense problem. A compact summary of the main LAPACK routines referenced throughout this chapter is collected in Appendix A.

3.1.3 Main Points of Comparison

The two flowcharts later in this chapter are intended as conceptual workflow diagrams rather than complete LAPACK call graphs. The main contrasts are:

- DSYEV spends most of its tridiagonal-eigenvector work inside a QR-style iteration with many fine-grained updates.
- DSYEVD reorganizes the same task into recursive subproblems, merge steps, and larger matrix operations.
- DSYEVD needs more temporary storage, but this is exactly what allows it to exploit blocked operations and modern BLAS implementations more effectively.

3.2 Key Implementation Blocks

3.2.1 SYTRD: Householder Tridiagonalization in LAPACK

The LAPACK routine DSYTRD reduces a dense symmetric matrix to

$$Q^T A Q = T,$$

where T is tridiagonal. This step is shared by both `DSYEV` and `DSYEVD`, so its main role in this chapter is to provide common implementation context before the two drivers diverge at the tridiagonal eigensolver stage.

In LAPACK, the reduction is carried out with Householder reflectors stored implicitly. The tridiagonal entries are written to the arrays `D` and `E`, while the reflector data are packed into `A` and `TAU`. This design avoids forming the dense orthogonal matrix Q immediately, allowing later routines such as `DORGTR` or `DORMTR` to reconstruct or apply it only when eigenvectors are required.

From the implementation perspective, the key point is that `DSYTRD` uses a blocked structure with two distinct performance regimes:

- **Panel processing (unblocked path):** Columns in the active panel are reduced one-by-one using Householder reflectors. This stage is carried out by routines such as `dlatrd` and `dsytd2`, which internally call Level-1/2 BLAS kernels (e.g., `dlarfg`, `daxpy`, `dsymv`, `dscal`) to construct and apply individual reflectors. This phase is memory-bound and typically limits parallel scalability.
- **Trailing matrix update (blocked path):** Once a block of reflectors has been formed, they are applied together to the unreduced trailing submatrix using the Level 3 BLAS kernel `dsyr2k`. This step achieves high arithmetic intensity and forms the performance-critical portion of the routine.

This split explains why `DSYTRD` is important but not decisive for the `DSYEV` versus `DSYEVD` comparison. Its total cost is still about $\frac{4}{3}n^3$ flops, and the panel phase remains limited by Level 2 BLAS behavior, so both drivers inherit a similar front-end cost. The main source of their later performance difference therefore lies not in `DSYTRD` itself, but in how the tridiagonal problem and eigenvectors are handled after this shared reduction.

3.2.2 *STEQR: Implicit QR/QL Iteration for Tridiagonal Matrices*

Overview and Function of DSTEQR

The LAPACK routine `DSTEQR` implements the implicit QR/QL method for computing all eigenvalues and optionally eigenvectors of a real symmetric *tridiagonal* matrix. In the dense symmetric driver `DSYEV`, it is used after reduction to tridiagonal form:

$$A \xrightarrow{\text{DSYTRD}} Q^\top A Q = T \xrightarrow{\text{DORGTR}} Q \xrightarrow{\text{DSTEQR('V')} \text{ with } Z=Q} U = Q U_T.$$

The implementation point that matters here is that `DSYEV` forms the dense basis Q first and then asks `DSTEQR('V')` to accumulate the tridiagonal rotations directly onto that basis.

Workflow in the DSYEV Path

For the present comparison, the essential point is that the `DSYEV` eigenvector path is built around an iterative QR/QL process on the tridiagonal matrix. Figure 3.1 summarizes this path at a conceptual level.

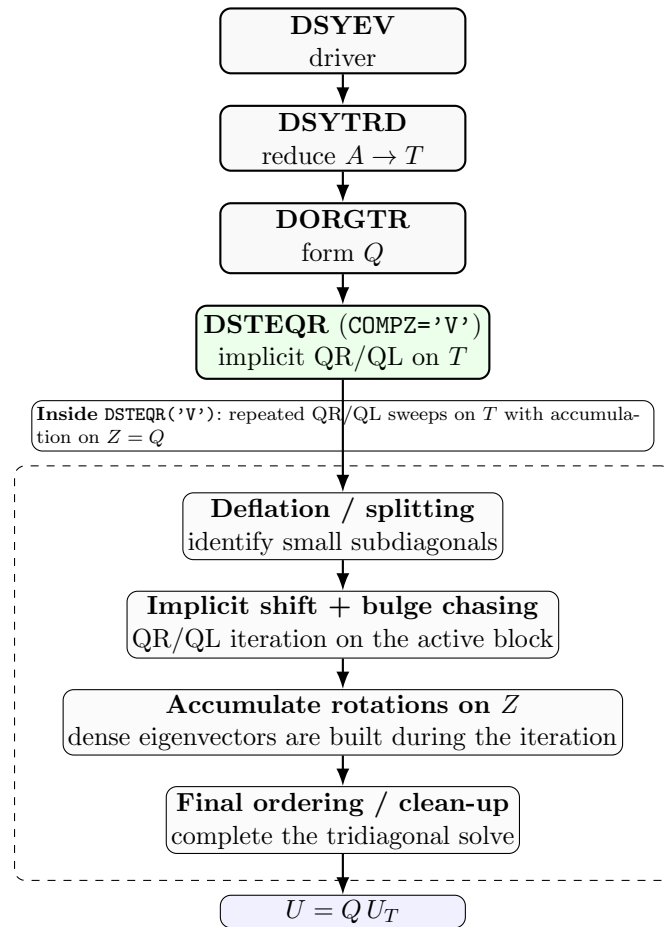


Figure 3.1: Conceptual workflow of the eigenvector path in DSYEV. The important feature is that eigenvector formation stays inside a QR/QL iteration built from many sequential updates.

Figure 3.1 emphasizes the main structural point: DSTEQR repeatedly updates a small active tridiagonal block and applies the same rotations to the dense eigenvector basis. The eigenvector path is therefore dominated by fine-grained sequential work rather than large blocked matrix operations.

Discussion: Eigenvector Path and Practical Implications

Compared to divide-and-conquer, DSTEQR has modest workspace requirements and a regular memory-access pattern. Its accumulation strategy, which applies QR/QL rotations directly to Z , also maintains orthogonality in a straightforward way. However, when many eigenvectors are required for large dense problems, the limited exposure to Level 3 BLAS usually leads to weaker performance and less parallel scalability than DSTEDC. The QR path is therefore best understood as the lighter-weight realization rather than the performance-oriented one.

3.2.3 STEDC: Divide-and-Conquer Implementation in LAPACK

Overview and Function of DSTEDC

The LAPACK routine DSTEDC implements the divide-and-conquer (D&C) method for computing all eigenvalues and eigenvectors of a real symmetric tridiagonal

matrix. In the dense symmetric driver `DSYEVD` with `JOBZ='V'`, LAPACK uses `DSTEDC('I')` to compute the eigenvectors U_T of the tridiagonal matrix T , and then applies the Householder reflectors by `DORMTR` to obtain the eigenvectors of the original dense matrix A . Although `DSTEDC` also supports `COMPZ='V'` in general, that is not the path used by `DSYEVD`.

In the driver routine `DSYEVD`, this step occurs after tridiagonalization:

$$A \xrightarrow{\text{SYTRD}} Q^T A Q = T \xrightarrow{\text{DSTEDC('I')}} U_T \xrightarrow{\text{DORMTR}} U = Q U_T.$$

The key implementation contrast with `DSYEV` is the ordering of work: `DSYEVD` first solves the tridiagonal problem in its own basis and only afterwards maps those vectors back to the dense basis through `DORMTR`.

Workflow in the `DSYEVD` Path

Within the `DSYEVD` driver path, the tridiagonal eigensolver is no longer an extended QR-style iteration. Instead, `DSTEDC('I')` works by splitting the tridiagonal problem into smaller subproblems, solving them, and merging them back together. Figure 3.2 summarizes this workflow at the level needed for the present thesis.

Figure 3.2 again emphasizes structure rather than a complete list of helper routines. What matters is that `DSTEDC` pushes more of the heavy work into split, merge, and blocked-update stages.

Discussion: Eigenvector Path and Performance Implications

Compared with the traditional QR iteration (`DSTEQR`), the D&C approach reduces both arithmetic complexity and bandwidth pressure in the eigenvector path. Inside `DSTEDC`, the dominant work is organized as blocked merge updates, which are well matched to Level-3 BLAS. The final dense-basis mapping is then carried out by `DORMTR`. Although `DSTEDC` requires additional workspace for temporary eigenvector blocks, this cost is compensated by reduced arithmetic and improved cache reuse, which usually yields better scalability for large dense matrices. In practice, the expensive merge path in LAPACK is funneled largely into `DGEMM`-dominated blocked work, which is exactly the implementation feature that later explains the performance advantage of `DSYEVD`.

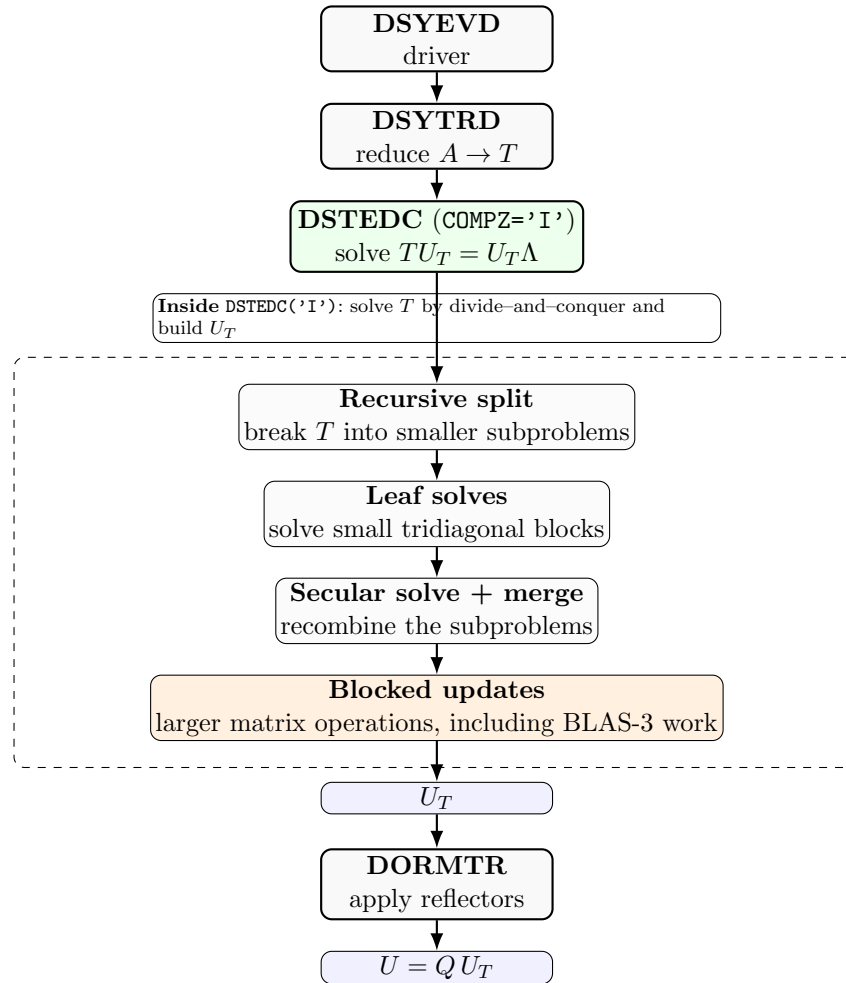


Figure 3.2: Conceptual workflow of the eigenvector path in DSYEVD. The important feature is that the tridiagonal problem is reorganized into recursive subproblems and blocked merge work before the final dense-basis mapping.

3.3 Implementation Implications

Chapter 2 presented the mathematical logic of the symmetric eigenvalue problem in three stages: reduction to tridiagonal form, solution of the tridiagonal eigenproblem, and recovery of eigenvectors for the original dense matrix. The LAPACK implementations preserve this broad structure, but the practical difference between DSYEV and DSYEVD appears in the organization of the second and third stages. The purpose of this section is to distill the implementation details discussed earlier into the specific structural differences that matter for later performance interpretation.

Table 3.2: Implementation-level comparison of the main stages in DSYEV and DSYEVD.

Stage	Driver realization	Why it matters
Dense $\rightarrow$ tridiagonal reduction	DSYTRD with Householder reflectors stored implicitly in A and TAU	Shared by both drivers; blocked updates expose some BLAS-3, but panel work remains memory-bound.
Tridiagonal eigensolution (QR path)	DSTEQR with deflation, implicit shifts, Givens rotations, and optional accumulation on Z	Small workspace and robust structure, but limited Level-3 BLAS usage reduces scalability for large problems.
Tridiagonal eigensolution (D&C path)	DSTEDC through the DLAED* family: splitting, secular solving, deflation, and merging	Requires more workspace, but blocked merge steps map well to DGEMM and modern hardware.
Back transformation to dense eigenvectors	DSYEV: DORGTR + DSTEQR('V'); DSYEVD: DORMTR after DSTEDC('I')	Same mathematical target $U = QU_T$, but different routine ordering and data flow.

Table 3.2 highlights the central implementation point of this chapter. The decisive difference between DSYEV and DSYEVD does not arise in the shared reduction routine DSYTRD; rather, it arises in the tridiagonal eigensolver and in the subsequent treatment of eigenvectors. The DSYEV path, based on DSTEQR, is dominated by sequential bulge-chasing updates and plane rotations. By contrast, the DSYEVD path, based on DSTEDC, reorganizes the same stage into recursive subproblems and blocked merge operations. This reorganization increases workspace requirements, but it also creates substantially more Level-3 BLAS work and correspondingly greater opportunity for cache reuse and parallel execution. From an implementation perspective, DSYEV is therefore the lighter-weight driver, whereas DSYEVD trades additional workspace for a more hardware-efficient structure. This is the bridge to Chapter 4: the empirical advantage of DSYEVD should be sought primarily in the tridiagonal eigensolver and eigenvector-handling stages, not in the shared dense tridiagonalization stage.

CHAPTER 4

Experimental Evaluation and Performance Analysis

This chapter evaluates the performance differences between the two LAPACK symmetric eigensolvers `DSYEV` (QR algorithm) and `DSYEVD` (divide-and-conquer), with emphasis on the case `JOBZ='V'` in which both eigenvalues and eigenvectors are required. Here, LAPACK uses `JOBZ='N'` for eigenvalues only and `JOBZ='V'` for both eigenvalues and eigenvectors. The central empirical objective is to determine not only which driver is faster on the test platform, but also which stage creates the runtime gap and how much of that gap is associated with algorithmic structure, optimized libraries, and SIMD width.

The chapter proceeds in five steps. Section 4.1 defines the hardware platform, software stack, test matrices, and measurement protocol. Section 4.2 first establishes the end-to-end result: `DSYEVD` is consistently faster than `DSYEV`, and the dominant difference appears in the tridiagonal eigensolver stage. Sections 4.3 and 4.4 then use controlled comparisons to separate three explanatory layers: algorithm choice (`DSTEQR` versus `DSTEDC`), the overall Netlib-to-OpenBLAS implementation change, and the incremental gain from 256-bit SVE over a 128-bit SIMD baseline. Section 4.5 then uses call-path profiling and hardware-counter evidence to explain the behavior observed in these controlled comparisons. Finally, Section 4.6 assembles the chapter-level synthesis and implications.

At the chapter level, the main conclusion is that `DSYEVD` outperforms `DSYEV` on this platform primarily because it replaces the QR-based tridiagonal eigensolver by `DSTEDC`. This advantage, however, is accompanied by substantially higher workspace demand.

4.1 Experimental Setup

All experiments were conducted on an AWS EC2 `c7g.xlarge` instance running Linux and equipped with an AWS Graviton3 ARM processor. The following sections summarize only the experimental context needed to read the benchmark results. Additional configuration details are collected in Appendix B.

4.1.1 Hardware Platform

Experiments were performed on an AWS EC2 `c7g.xlarge` instance equipped with an AWS Graviton3 processor (ARMv8-A with SVE support). This distinction matters for the controlled comparison in Section 4.4: the processor supports both the conventional 128-bit ARM SIMD extension (ASIMD/NEON) and the newer SVE vector extension. On this system, Linux reports an SVE default vector length of 32 bytes, that is, 256 bits. The system characteristics are:

- **CPU:** 4 vCPUs, single hardware thread per core

- **Memory:** 8 GiB RAM

All controlled comparisons were run in single-threaded mode to isolate algorithmic and library-level performance differences. The overall benchmark in Section 4.2 additionally includes a small multithread scaling check with 1, 2, and 4 threads on the same machine.

4.1.2 Software Configuration

LAPACK and BLAS Libraries. The experiments use three library configurations, based on Netlib LAPACK and reference BLAS [2, 12, 13] and on OpenBLAS [14]:

Table 4.1: Library configurations for the current DSYEV/DSYEVD experiments.

Configuration	LAPACK layer	BLAS implementation
LAPACK	Reference LAPACK	Reference BLAS
OpenBLAS SIMD	Reference LAPACK	OpenBLAS (SIMD)
OpenBLAS SVE	Reference LAPACK	OpenBLAS (SVE)

The `OpenBLAS SIMD` configuration is built for the generic `ARMV8` target without SVE and still uses 128-bit ASIMD/NEON instructions in kernels such as `DGEMM`. The `OpenBLAS SVE` configuration is built for the Graviton3 SVE-capable target (`NEOVERSEV1`). For the benchmark runs in Section 4.2, the active OpenBLAS SVE library is a pthread build with `MAX_THREADS=8`; since the test machine has 4 vCPUs, the thread scaling check is limited to 1, 2, and 4 threads.

For the DSYEV/DSYEVD experiments, all three configurations follow the same Reference LAPACK lineage at the LAPACK layer. In particular, the OpenBLAS SIMD and OpenBLAS SVE builds use the same LAPACK-side routines for DSYEV, DSYEVD, DSYTRD, DSTEQR, DSTEDC, DORGTR, and DORMTR; they differ in the underlying BLAS kernels. OpenBLAS also includes additional integration code and some optimized replacements for other routines, but those are not material to the present DSYEV/DSYEVD study. Exact compiler flags, build targets, and thread-control settings are listed in Appendix B.

Performance Measurement. Timing measurements were obtained using wrapper functions that instrument LAPACK routines at the subroutine level. This allows us to break down the total runtime into:

- Tridiagonal reduction: DSYTRD
- Tridiagonal eigensolver: DSTEQR (for DSYEV) or DSTEDC (for DSYEVD)
- Eigenvector back-transformation: DORGTR (for DSYEV) or DORMTR (for DSYEVD)

The benchmark and controlled-comparison tables in Sections 4.2–4.4 use the same matrix sizes, the same stage-level wrapper scheme, and repeated runs under a fixed threading configuration. The deeper profiling and hardware-counter experiments use heavier instrumentation and are therefore interpreted primarily as explanatory evidence rather than as exact timing replicas. The full run protocol, thread-control settings, and correctness checks are summarized in Appendix B. For reproducibility, the accompanying repository [15] provides the scripts, code, and result snapshots corresponding to the experiments reported in this chapter.

4.1.3 Test Matrices

All experiments use Symmetric Positive Definite (SPD) Kac–Murdock–Szegő (KMS) matrices with correlation parameter $\rho = 0.95$ [16]. A KMS matrix has entries

$$A_{ij} = \rho^{|i-j|}, \quad 0 < \rho < 1,$$

so it is a dense symmetric Toeplitz matrix whose off-diagonal entries decay smoothly with distance from the diagonal.

This matrix family is convenient for the present experiments for three reasons. First, it is guaranteed to be symmetric positive definite for $0 < \rho < 1$, so both `DSYEV` and `DSYEVD` can be tested on a stable and well-defined class of inputs. Second, the matrix is fully dense, so the experiment exercises the dense reduction, tridiagonal eigensolver, and back-transformation stages that these LAPACK drivers are designed for, without introducing sparsity-specific effects. Third, the single parameter ρ provides a simple way to control how strongly coupled the rows and columns are: choosing $\rho = 0.95$ keeps the matrix far from diagonal, so the eigenvector computation remains nontrivial while the family itself stays easy to describe and reproduce.

Matrix sizes tested:

$$n \in \{512, 1024, 2048, 4096\}.$$

Appendix B gives the detailed matrix specification, illustrative small example, and the residual tests used to validate numerical correctness.

4.2 Overall Performance Benchmark

This section presents the overall performance comparison between `DSYEV` and `DSYEVD` using the optimized OpenBLAS SVE build. This benchmark section therefore measures the SVE-enabled OpenBLAS configuration, not the 128-bit OpenBLAS SIMD baseline used later in Section 4.4. We first report the single-thread baseline and then a small thread-scalability check over 1, 2, and 4 threads. We also report the phase-level runtime breakdown defined in Section 4.1: tridiagonal reduction, tridiagonal eigensolution, and eigenvector back-transformation. The later profiling section uses heavier instrumentation, so small differences in absolute times are expected.

4.2.1 Single-Thread Performance

Table 4.2: Single-thread absolute runtime breakdown for DSYEV and DSYEVD (JOBZ='V', OpenBLAS SVE, median over repeated runs).

n	Routine	Total (s)	Tridiag. (DSYTRD) (s)	Tri-eig (DSTEQR/DSTEDC) (s)	Back-trans. (DORGTR, DORMTR) (s)
512	DSYEV	0.126	0.013	0.103	0.009
	DSYEVD	0.036	0.013	0.010	0.013
1024	DSYEV	0.932	0.095	0.772	0.064
	DSYEVD	0.239	0.095	0.050	0.094
2048	DSYEV	7.743	0.803	6.439	0.497
	DSYEVD	1.859	0.807	0.294	0.754
4096	DSYEV	66.255	6.574	55.793	3.853
	DSYEVD	14.482	6.616	1.935	5.914

Analysis. Across all sizes, DSYEVD is consistently faster than DSYEV. The main difference comes from the tridiagonal eigensolver stage. At $n = 4096$, DSTEQR accounts for about 84.2% of DSYEV runtime, whereas DSTEDC contributes only about 13.4% to DSYEVD. By contrast, the dense tridiagonalization stage is nearly the same for the two drivers because both call DSYTRD. For the full eigenpair case on this platform, DSYEVD is already the faster dense driver once $n \geq 1024$, and the gap widens with problem size. A necessary qualification, developed further in Section 4.6, is that DSYEVD attains this speed advantage at the cost of substantially higher workspace demand than DSYEV.

4.2.2 Thread Scalability

Table 4.3: Absolute runtime with 1, 2, and 4 threads (OpenBLAS SVE, JOBZ='V', median over repeated runs).

n	Routine	1 Thread [s]	2 Threads [s]	4 Threads [s]
512	DSYEV	0.126	0.118	0.115
	DSYEVD	0.036	0.024	0.017
1024	DSYEV	0.932	0.862	0.830
	DSYEVD	0.239	0.145	0.092
2048	DSYEV	7.743	7.218	6.892
	DSYEVD	1.859	1.106	0.657
4096	DSYEV	66.255	61.823	59.221
	DSYEVD	14.482	8.308	4.773

Table 4.4: Thread speedup and parallel efficiency at $n = 4096$ (OpenBLAS SVE, JOBZ='V'). Efficiency is speedup divided by thread count.

Routine	2 Threads		4 Threads	
	Speedup	Efficiency	Speedup	Efficiency
DSYEV	1.07	0.54	1.12	0.28
DSYEVD	1.74	0.87	3.03	0.76

Interpretation. For DSYEV, multithread scaling is very limited: even at $n = 4096$, the speedup is only $1.07\times$ from 1 to 2 threads and $1.12\times$ from 1 to 4 threads. In contrast, DSYEVD reaches $1.74\times$ and $3.03\times$ over the same thread counts, so its runtime advantage becomes larger on multicore systems. These thread results are secondary confirmation of the single-thread ordering, but they reinforce the same structural point: DSYEVD exposes substantially more useful parallel work than DSYEV.

4.3 Impact of the Netlib-to-OpenBLAS Implementation Change

This section compares the *Netlib* implementation (Netlib LAPACK + reference BLAS [2, 12, 13]) with the *OpenBLAS SVE* implementation [14]. Because both the LAPACK and BLAS layers change, this is not a BLAS-only comparison. Its purpose is to identify which stages benefit when the eigensolver moves from an unoptimized reference implementation to an optimized OpenBLAS implementation.

4.3.1 Stage-by-Stage Comparison

Table 4.5 presents the speedup achieved by the OpenBLAS SVE implementation over the Netlib implementation for each computational stage. Speedup is computed as the ratio of Netlib median time to OpenBLAS-SVE median time.

Table 4.5: Speedup of the OpenBLAS SVE implementation over the Netlib implementation by computational stage (single-threaded, speedup = Netlib time / OpenBLAS SVE time).

n	DSYTRD (reduction)	DSTEQR (Tri-eig)	DSTEDC (Tri-eig)	DORGTR (back-trans)	DORMTR (back-trans)
512	2.93	1.00	2.60	5.89	6.37
1024	2.94	1.00	3.48	6.83	7.02
2048	2.76	1.00	4.21	7.21	7.39
4096	2.77	1.00	5.08	7.67	7.77

Interpretation. Table 4.5 reveals a clear separation among stages. DSYTRD improves by about $2.8\text{--}2.9\times$, DSTEDC by $2.6\text{--}5.1\times$, and the back-transformation stages by $5.9\text{--}7.8\times$, whereas DSTEQR remains at $\approx 1.0\times$ on this platform. This pattern is consistent with the later profiles: DSTEQR spends most of its time in fine-grained rotation updates, whereas DSTEDC, DORGTR, and DORMTR expose blocked DGEMM-dominated work that optimized libraries accelerate well. The Netlib-to-OpenBLAS implementation change therefore benefits DSYEVD much more strongly than DSYEV.

A cleaner same-LAPACK comparison, in which only the kernel target changes, is presented next in Section 4.4.

4.4 Impact of SVE Vectorization

This section evaluates the impact of SVE vectorization by comparing a non-SVE OpenBLAS SIMD baseline [14] built for the generic ARMV8 target against an OpenBLAS build with SVE support. On this platform, the baseline already uses 128-bit ASIMD/NEON kernels, so the comparison measures the incremental benefit of moving from that vectorized baseline to a wider 256-bit SVE implementation. In principle, doubling the vector width from 128 bits to 256 bits can increase the throughput of well-vectorized floating-point kernels, because each vector instruction can operate on twice as many double-precision elements. That upper bound is not generally achieved in practice, however, because realized speedup also depends on arithmetic intensity, memory behavior, instruction mix, and how much of the routine runtime is actually spent inside width-sensitive vector kernels.

4.4.1 Stage-by-Stage SIMD Speedup

Table 4.6 presents the speedup achieved by the OpenBLAS SVE build over the OpenBLAS SIMD baseline for each computational stage. Speedup is computed from repeated-run medians under the same single-threaded wrapper configuration.

Table 4.6: Speedup of the OpenBLAS SVE build over the 128-bit OpenBLAS SIMD baseline by stage (single-threaded, speedup = SIMD time / SVE time).

n	DSYTRD (syev)	DSTEQR (Tri-eig)	DSYTRD (syevd)	DSTEDC (Tri-eig)
512	1.59	1.01	1.59	1.07
1024	1.66	1.00	1.71	1.12
2048	1.29	1.00	1.49	1.16
4096	1.43	1.00	1.57	1.18

Interpretation. In principle, if a stage is already dominated by compute-bound vector kernels, then moving from 128-bit SIMD to 256-bit SVE should increase throughput. Table 4.6 shows that the measured behavior is more moderate. In the present results, DSYTRD gains about 1.3–1.7 $\times$ and DSTEDC about 1.07–1.18 $\times$, while DSTEQR remains at $\approx 1.0\times$. Thus, the speedup is clearly nonzero for some blocked stages, but it falls well short of the naive 2 $\times$ expectation from vector-width doubling.

This outcome indicates that vector width is only one component of performance. Once the OpenBLAS LAPACK layer is fixed, moving from 128-bit SIMD to SVE benefits mainly those stages whose inner kernels already vectorize well. DSTEQR shows essentially no gain because its fine-grained, dependency-heavy structure offers little useful SIMD throughput to begin with. DSYTRD benefits more because it contains more regular blocked kernels, whereas DSTEDC benefits only modestly because the non-SVE OpenBLAS baseline already executes its DGEMM-dominated merge path efficiently.

Taken together, these results indicate that the contribution of SVE in this chapter is smaller than that of the Netlib-to-OpenBLAS implementation change. The move from the Netlib implementation to optimized OpenBLAS produces the larger performance change, whereas the move from the 128-bit OpenBLAS SIMD baseline to OpenBLAS SVE yields an additional but comparatively modest gain. This pattern is consistent with the fact that enabling SVE improves the throughput of the same blocked kernel family without altering the underlying algorithmic or blocked structure of DSTEDC.

4.5 Detailed Performance Profiling

This section provides detailed performance profiling using instrumentation and profiling tools to understand where time is spent within each computational stage. The analysis clarifies the structural basis on which DSTEDC outperforms DSTEQR.

Unless otherwise noted, the percentages in this section come from dedicated profiling runs with finer-grained wrappers, call tracing, and associated logging overhead. They should therefore be read as profile-specific snapshots: small differences relative to the benchmark tables in Section 4.2 are expected and do not change the qualitative conclusions.

Table 4.7: Compact summary of the dominant profiled call paths at $n = 4096$.

Driver path	Stage	Dominant internal path	Share of end-to-end runtime
DSYEV	DSTEQR	DLASR	76.5%
DSYEV	DORGTR	DLARFB→DGEMM	6.0%
DSYEV	DSTEDC	DLAEDO→DGEMM	10.9%
DSYEV	DORMTR	DORMQL→DLARFB→DGEMM	38.3%

The detailed inclusive call trees for these four stages are moved to Appendix B.3; the main text keeps only the quantitative points needed for the performance argument.

4.5.1 Analysis of DSTEQR

Consistent with the benchmark trend in Section 4.2, DSYEV shows that the tridiagonal eigensolve DSTEQR dominates the runtime. In the profiling run highlighted here, at $n = 4096$, DSTEQR accounts for 82.7% of the total time, followed by the explicit back-transformation DORGTR (6.36%) and the tridiagonal reduction DSYTRD (10.9%).

Building on that pipeline view, we now instrument the major stages (`JOBZ='V'`, dense SPD KMS input [16]) that differentiate DSYEV from DSYEVD. Top-level timings are reported exclusively per stage, while the breakdowns adopt hierarchical (inclusive) views to reveal dominant call paths.

Focused call flow of DSTEQR. Percentages reported on nodes are shares of the *end-to-end* DSYEV time; the subtree is inclusive.

Key quantitative points. Within DSTEQR, DLASR alone accounts for 55.6440 s, that is, 92.5% of DSTEQR time and 76.5% of end-to-end DSYEV time. By contrast, DLARTG contributes less than 1% of stage time despite being called more than 12.7 million times, and the remaining named helpers are negligible.

Focused call flow of DORGTR. Percentages are shares of the DSYEV end-to-end runtime; the view remains inclusive.

Key quantitative points. The same concentration appears in DORGTR. The DLARFB→DGEMM path accounts for almost the whole stage, with DGEMM alone contributing 4.3667 s, or 94.4% of DORGTR and 6.00% of end-to-end DSYEV. This confirms that the back-transformation path is already efficiently blocked and is not the main leverage point for improving DSYEV.

Summary. The profiling evidence isolates the bottleneck in the DSYEV path: DSTEQR is overwhelmingly dominated by DLASR, whereas DORGTR is already dispatched through efficient blocked kernels (DLARFB to DGEMM) and contributes only a small share of the end-to-end runtime. Improving the back-transformation alone therefore cannot materially change the overall behavior of DSYEV.

4.5.2 Analysis of DSTEDC

Consistent with the benchmark trend in Section 4.2, DSYEVD shows a different stage-level distribution compared to DSYEV. In the profiling run highlighted here, at $n = 4096$, the tridiagonal reduction DSYTRD dominates (45.9%), followed by the back-transformation DORMTR (40.4%) and the tridiagonal eigen-solve DSTEDC (13.5%). This contrasts sharply with DSYEV, where DSTEQR accounts for 82.7% of the runtime.

Building on that pipeline view, we now instrument the major stages (JOBZ='V', dense SPD KMS input [16]) to reveal their internal structure. Top-level timings are reported exclusively per stage, while the breakdowns of DSTEDC and DORMTR adopt hierarchical (inclusive) views to highlight dominant call paths and sources of computational intensity.

Focused call flow of DSTEDC. Percentages on nodes are shares of *end-to-end* DSYEVD time; the DSTEDC subtree is inclusive by design.

Key quantitative points. Within DSTEDC, DLAED0 accounts for 98.4% of stage time, and the merge chain DLAED1/DLAED3 accounts for almost all of that on an inclusive basis. Most importantly, DGEMM alone contributes 1.7606 s, which is 80.6% of DSTEDC time and 10.9% of end-to-end DSYEVD time. By contrast, the remaining named routines inside DSTEDC are individually small, with the next largest term, DLAED4, contributing only 9.4% of stage time.

Focused call flow of DORMTR. We similarly expose the back-transform path; percentages are shares of end-to-end DSYEVD, inclusive within this subtree.

Key quantitative points. The same concentration appears in DORMTR. The DORMQL→DLARFB→DGEMM path is effectively the whole stage, and DGEMM alone contributes 6.1812 s, which is 94.7% of DORMTR time and 38.3% of end-to-end DSYEVD time. The next largest named component, DTRMM, contributes only 2.2% of DORMTR time.

Summary. The profiling evidence for DSYEVD is structurally different from that of DSYEV. Inside DSTEDC, most of the heavy work is funneled through the merge chain and ultimately through DGEMM, while the DORMTR back-transformation is likewise dominated by DLARFB→DGEMM. The remaining non-DGEMM components are individually small, so the main runtime leverage in the DSYEVD path is concentrated in blocked kernels rather than in fine-grained scalar updates. Mechanistically, this happens because DSTEDC merges subproblems by applying eigenvector updates in

block form, thereby exposing large DGEMM-dominated work. The detailed trees are collected in Appendix B.3; the main conclusion for the present chapter is simply that the fast DSYEVD path is DGEMM-dominated, whereas the slow DSYEV path is dominated by fine-grained rotation work.

4.5.3 Hardware-Efficiency Evidence

To complement the call-path profiling in Sections 4.5.1 and 4.5.2, we collected direct hardware-counter and memory measurements under the same experimental configuration used in the earlier sections. For small sizes we repeated runs multiple times and report the average per run; for large sizes a single run was used to keep the measurement cost manageable. Since the runtime comparison between DSYEV and DSYEVD has already been established in Section 4.2, the emphasis here is not on repeating the timing comparison, but on addressing a narrower follow-up to Section 4.5.2. Section 4.5.2 identified the fast path as DGEMM-dominated blocked work; the present section asks whether that path also reduces the total hardware cost. The evidence below therefore focuses on cumulative work and memory cost rather than on attributing performance to any one micro-mechanism inside DGEMM.

Table 4.8: End-to-end memory footprint of DSYEV and DSYEVD. Peak RSS is reported in MiB.

n	DSYEV RSS [MiB]	DSYEVD RSS [MiB]	Ratio
512	6.6	11.1	$1.69\times$
1024	19.0	35.9	$1.89\times$
2048	67.6	134.1	$1.98\times$
4096	260.6	521.6	$2.00\times$

Table 4.8 isolates the end-to-end memory tradeoff. The faster DSYEVD path consistently requires more workspace, and the ratio approaches $2\times$ for large matrices. This is fully consistent with the design of divide-and-conquer: it accelerates the eigensolve at the cost of substantially larger temporary storage.

Table 4.9: Compact hardware evidence for the isolated tridiagonal eigensolver stage. Instruction counts and backend-memory stall counts are reported per run.

n	DSTEQR instr. [$\times 10^9$]	DSTEDC instr. [$\times 10^9$]	Instr. ratio	DSTEQR stalls [$\times 10^9$]	DSTEDC stalls [$\times 10^9$]	Stall ratio
2048	73.52	3.12	$23.60\times$	0.717	0.118	$6.07\times$
4096	602.34	40.28	$14.95\times$	6.043	2.163	$2.79\times$

Table 4.9 reports the most informative counter-level evidence for the isolated tridiagonal eigensolver stage, DSTEQR executes far more instructions than DSTEDC: about $23.6\times$ more at $n = 2048$, $15.0\times$ more at $n = 4096$, respectively. It also accumulates substantially more backend-memory stalls in absolute terms, by factors of $6.1\times$ and $2.8\times$, respectively. These are cumulative costs over the full stage and therefore match the qualitative picture from Sections 4.5.1 and 4.5.2: the implicit QR route spends much more total effort in fine-grained updates, whereas the divide-and-conquer route concentrates more work into fewer, denser kernels. This is the

main evidential role of Section 4.5.3: not to establish the internal causes of **DGEMM** performance in isolation, but to show that the **DGEMM**-dominated **DSTEDC** path indeed carries a much smaller total hardware cost than the **DSTEQR** path.

Normalized counters such as IPC or refill-per-instruction are therefore not the right headline metrics here. **DSTEDC** can exhibit substantial cache traffic because **DGEMM** is a high-throughput streaming kernel, yet still finish far earlier because the total amount of work is so much smaller. The stronger conclusion is therefore based on cumulative instruction count and cumulative backend-memory stall cost, not on any single normalized counter in isolation.

For completeness, Tables 4.10 and 4.11 report two normalized deep-counter views of the same isolated tri-eig stage.

Table 4.10: Deep-counter view of the isolated tridiagonal eigensolver: load/store instruction mix. Shares are measured relative to total instructions.

n	DSTEQR	DSTEDC	DSTEQR	DSTEDC
	load share	load share	store share	store share
1024	0.273	0.274	0.180	0.053
2048	0.273	0.295	0.179	0.037
4096	0.274	0.306	0.175	0.033

Table 4.11: Deep-counter view of the isolated tridiagonal eigensolver: L1 load-miss rate and backend-memory stalls per instruction.

n	DSTEQR	DSTEDC	DSTEQR	DSTEDC
	L1 miss rate	L1 miss rate	stall / inst	stall / inst
1024	0.0044	0.0338	0.0069	0.0266
2048	0.0047	0.0783	0.0083	0.0378
4096	0.0188	0.1230	0.0100	0.0578

These deeper counters refine the interpretation but do not overturn it. Relative to **DSTEQR**, **DSTEDC** shows a slightly higher load share and a much lower store share in Table 4.10, which is consistent with a denser, more streaming-oriented kernel structure. At the same time, Table 4.11 shows that **DSTEDC** has the higher L1 miss rate and the higher stall-per-instruction value on this platform. This is not a contradiction: a **DGEMM**-dominated kernel can stream data aggressively and still finish much sooner because the total instruction count and total stall count are far smaller. The cumulative counters in Table 4.9 therefore remain the primary evidence, while the deeper tables mainly support the interpretation of the operation mix.

Overall, the hardware-counter evidence supports the earlier structural interpretation. The main advantage of **DSTEDC** is that the tri-eig stage does much less total work and organizes that work in a more hardware-friendly form. As a result, **DSYEVD** is faster than **DSYEV** because **DSTEDC** executes far fewer instructions, accumulates less total memory-stall cost, and follows the same blocked-**DGEMM**-dominated structure

exposed by the call-path profiles in Section 4.5.2. The price is a significantly larger workspace, visible in the roughly $2\times$ increase in peak memory usage for `DSYEVD` in Table 4.8. Read together, Sections 4.5.2 and 4.5.3 therefore establish a clean two-step explanation: first identify that the fast `DSYEVD` route is `DGEMM`-dominated blocked work, then show that this route also incurs far less total hardware work than the rotation-dominated `DSYEV` path.

4.6 Final Synthesis and Implications

This final section synthesizes the earlier benchmark, controlled-comparison, profiling, and hardware-counter results. Its role is to assemble the chapter’s main explanatory argument and overall implications rather than to introduce a separate experiment.

Synthesis. `DSTEQR` concentrates its work in fine-grained rotation updates, most of it through `DLASR`. That structure is hard to accelerate with optimized BLAS libraries and offers little useful thread- or SIMD-level parallelism. `DSTEDC`, by contrast, channels most of its heavy work into blocked merge operations dominated by `DGEMM`. This is the kind of work that optimized BLAS implementations and wider vectors handle well. The profiling and hardware-counter sections therefore do not introduce a new explanation; they validate the attribution made earlier by showing the concrete kernel paths that implement it.

The controlled comparisons earlier in the chapter support the same conclusion from three complementary angles. First, with the same OpenBLAS SVE backend, the tridiagonal eigensolver stage shows a $10.38\times$ to $28.88\times$ advantage for `DSTEDC` over `DSTEQR`, so the algorithmic reorganization sets the scale of the gap. Second, the Netlib-to-OpenBLAS implementation comparison shows that the blocked stages gain strongly from the optimized implementation, whereas `DSTEQR` remains close to $1.0\times$. Third, the 128-bit SIMD to 256-bit SVE comparison shows only an incremental improvement, with `DSTEDC` gaining $1.07\text{--}1.18\times$ and `DSYTRD` $1.3\text{--}1.7\times$. The ordering of importance is therefore consistent throughout the chapter: algorithmic restructuring first, optimized BLAS implementation second, and wider vectors third.

Implications. For the full eigenpair case studied here, `DSYEVD` is the faster dense symmetric eigensolver on this platform once the problem is at least moderately large; in the present measurements, this ordering is already clear by $n \geq 1024$. The difference is driven by the substantially faster tridiagonal eigensolver stage, whereas the corresponding cost is higher memory demand: the divide-and-conquer path requires substantially larger workspace, approaching roughly twice the peak RSS of `DSYEV` in the end-to-end measurements. In this sense, `DSYEV` remains of interest primarily in settings where workspace pressure is more restrictive than runtime.

At the end-to-end level, the attribution is correspondingly narrow: essentially all net runtime saving is created by the tridiagonal eigensolver stage, while other stages partly offset rather than create that gain. The choice between the two drivers is therefore governed mainly by the tradeoff between a much faster tri-eig path and substantially higher workspace demand.

Table 4.12: Summary comparison between DSYEV and DSYEVD for the full eigenpair case, with representative $n = 4096$ runtime and memory values.

Aspect	DSYEV	DSYEVD
Tri-eig routine	DSTEQR (implicit QR iteration)	DSTEDC (divide-and-conquer)
Main work pattern	Rotation-heavy updates that are hard to accelerate	Blocked merge/update path dominated by DGEMM
Total runtime at $n = 4096$	66.255 s	14.482 s
Peak memory at $n = 4096$	260.6 MiB	521.6 MiB
Primary advantage	Lower workspace demand	Much faster full eigenpair computation
Primary limitation	Tri-eig stage dominates the end-to-end cost	Roughly $2\times$ higher peak memory
More appropriate when	Memory pressure matters more than runtime	Runtime matters and workspace is available

Runtime values come from Section 4.2. Peak memory comes from Section 4.5.3.

Table 4.12 summarizes the main tradeoff of this chapter in compact form. DSYEV and DSYEVD mainly differ in the tridiagonal eigensolver they use, and that design choice propagates to the hot path, total runtime, and memory demand. At $n = 4096$, DSYEVD is $4.57\times$ faster overall, while its peak RSS is approximately $2\times$ higher. The detailed instruction and stall-count evidence remains in Section 4.5.3; the table here is limited to the quantities most directly relevant to the comparison between the two drivers.

The empirical conclusions of this chapter are intentionally scoped to the KMS matrix family, the tested ARM platform, and the library builds centered on OpenBLAS. Within that scope, the advantage of DSTEDC over DSTEQR is associated with a reorganization of the dominant work into blocked kernels that the software and hardware stack execute efficiently.

CHAPTER 5

Conclusion and Future Work

5.1 Main Findings

This thesis compared `DSYEV` and `DSYEVD` through three linked layers: the mathematical structure of their tridiagonal eigensolvers, their realization in LAPACK, and phase-level performance experiments. The central conclusion is that the practical advantage of `DSYEVD` over `DSYEV` is driven primarily by algorithmic structure rather than by micro-optimization alone.

Both drivers share the same high-level pipeline: reduction to tridiagonal form, solution of the tridiagonal eigenproblem, and back-transformation of eigenvectors when required. The decisive difference appears in the tridiagonal stage: `DSYEV` uses `DSTEQR` (QR iteration), while `DSYEVD` uses `DSTEDC` (divide-and-conquer). Experimental results in Chapter 4 show that this change yields the dominant speed benefit, with tri-eig stage speedup on the order of $\sim 15\times$ on average. The dense reduction stage behaves similarly for the two drivers, which is consistent with the fact that both use the same `DSYTRD` preprocessing step. The final eigenvector-recovery step serves the same mathematical purpose in both cases, namely forming $U = Q U_T$, but it is implemented through different routine paths.

A second conclusion is that library-level optimization amplifies, but does not create, the main speedup. `DSTEDC` benefits substantially from BLAS-3 kernels, especially `DGEMM`, and therefore gains from optimized BLAS implementations. In contrast, `DSTEQR` remains dominated by finer-grained BLAS-1/2 style operations and receives much less benefit from the same backend improvements.

A third conclusion is that routine selection should be interpreted as a runtime–workspace tradeoff rather than as a universal ranking. In the full eigenpair setting studied here, `DSYEVD` is preferable when runtime is the main concern, whereas `DSYEV` remains relevant when workspace is more constrained or when problem sizes are small enough that absolute runtime differences are modest.

Overall, the evidence supports a clear hierarchy of impact:

1. Algorithmic choice at the tridiagonal stage is dominant.
2. BLAS implementation quality is important but secondary.
3. Practical routine selection reflects a runtime–workspace tradeoff.

5.2 Practical Guidance for Routine Selection

Based on the theoretical and experimental results of this thesis, choice between `DSYEV` and `DSYEVD` should be guided by output requirements, matrix size, and resource constraints.

When both eigenvalues and eigenvectors are required, DSYEVD is the preferable choice in the tested setting when runtime is the main concern. Its divide-and-conquer tridiagonal solver and BLAS-3-friendly structure provide substantially lower runtime for medium-to-large matrices. This advantage typically becomes more pronounced as problem size increases and as hardware parallelism is available.

When only eigenvalues are required, the practical timing difference between the two drivers is often small in the tested setting. This is consistent with expectation: both drivers share the same dense reduction and neither path performs dense eigenvector recovery. In this case, either routine can be acceptable, and selection may depend on existing code path consistency or implementation convenience.

DSYEV can still be reasonable in constrained scenarios, for example when memory usage and implementation simplicity are prioritized, or when matrix dimensions are small enough that absolute runtime differences are not critical for the application workflow.

Any fair comparison between the two routines should control thread counts, matrix families, problem sizes, and library configuration so that the measured differences reflect algorithmic behavior rather than accidental environment effects.

5.3 Limitations of This Study

Although the conclusions are consistent across the presented experiments, this study has several scope limitations.

First, the performance evaluation is centered on a specific software-hardware stack, with emphasis on LAPACK routines backed by OpenBLAS on the tested CPU platform. Results may vary across other vendor libraries and architectures, especially where kernel implementations and threading strategies differ.

Second, the matrix set and size range are intentionally focused to support a clear phase-level comparison between DSYEV and DSYEVD. While this design improves interpretability, it does not cover all matrix structures encountered in scientific applications.

Third, this thesis concentrates on single-node CPU behavior. GPU acceleration, heterogeneous CPU-GPU workflows, and distributed-memory multi-node scalability are outside the present scope.

Fourth, the analysis emphasizes runtime decomposition and implementation mechanisms. A broader study could include a more extensive joint treatment of accuracy, robustness, memory footprint, and energy efficiency under varied workloads.

Fifth, although workspace requirements are part of the practical tradeoff between DSYEV and DSYEVD, this thesis did not carry out a systematic cross-platform study of runtime and workspace consumption together. The conclusions about routine selection should therefore be read primarily as runtime-oriented guidance within the tested setting.

These limitations do not change the main conclusion of this thesis, but they define the boundary within which the reported quantitative results should be interpreted.

5.4 Future Work

Several directions can extend this work and strengthen its practical impact.

A first direction is broader validation across libraries and architectures, including OpenBLAS, MKL, BLIS, and vendor-specific implementations under a unified measurement protocol. This would help separate algorithmic effects from library-specific engineering choices more systematically.

A second direction is to broaden the workload model. Future experiments can include wider matrix families, different spectral structures, and varying conditioning, so that the runtime–stability tradeoff can be assessed beyond the present focused benchmark setting.

A third direction is to extend the evaluation beyond runtime alone. A joint study of runtime, numerical accuracy, workspace consumption, and possibly energy usage would provide stronger guidance for routine selection in realistic scientific computing environments. The same framework could then be extended to GPU-enabled, heterogeneous, and distributed settings.

APPENDIX A

Summary of Main LAPACK Routines

This appendix collects the small set of LAPACK routines that appear repeatedly in the implementation discussion of Chapters 2 and 3. The purpose is not to replicate LAPACK documentation, but to record the role each routine plays in the present thesis.

Table A.1: Main LAPACK routines discussed in the thesis.

Routine	Role in this thesis	Used in
DSYTRD	Reduces a dense symmetric matrix to tridiagonal form and stores the Householder reflectors needed for later eigenvector reconstruction.	DSYEV, DSYEVD
DSTEQR	Solves the symmetric tridiagonal eigenproblem by implicit QR/QL iteration; in the DSYEV path it also updates eigenvectors through many sequential rotations.	DSYEV
DSTEDC	Solves the symmetric tridiagonal eigenproblem by divide-and-conquer, organizing the dominant work into recursive subproblems and blocked merge operations.	DSYEVD
DORGTR	Forms the orthogonal matrix Q explicitly from the reflectors stored during DSYTRD.	DSYEV
DORMTR	Applies the stored orthogonal transformations to map tridiagonal eigenvectors back to the eigenvectors of the original dense matrix.	DSYEVD

APPENDIX B

Detailed Experimental Setup

This appendix records the experimental details that support the main results in Chapter 4 but are not essential to keep in the chapter’s main flow.

B.1 Hardware and Software Configuration

All experiments were run on an AWS EC2 `c7g.xlarge` instance equipped with an AWS Graviton3 processor (ARMv8-A with SVE support). Linux reports a default SVE vector length of 32 bytes, that is, 256 bits. The machine characteristics are:

- **CPU:** 4 vCPUs, single hardware thread per core
- **Memory:** 8 GiB RAM
- **Cache hierarchy:** 64 KiB L1 data cache and 64 KiB L1 instruction cache per core, 1 MiB L2 cache per core, and a shared 32 MiB L3 cache

All code was compiled using GCC with the following optimization flags:

```
-O3 -std=c11 -mcpu=native -mtune=native
-fno-math-errno -fno-trapping-math -ffp-contract=fast
```

Chapter 4 compares three library configurations: Reference LAPACK with Reference BLAS, Reference LAPACK with OpenBLAS SIMD, and Reference LAPACK with OpenBLAS SVE; see Table 4.1. The `OpenBLAS SIMD` build targets generic ARMV8 kernels without SVE, whereas the `OpenBLAS SVE` build targets the Graviton3-capable `NEOVERSEV1` backend. For the benchmark runs in Section 4.2, the active OpenBLAS SVE library is a pthread build with `MAX_THREADS=8`. Since the machine has 4 vCPUs, the explicit thread-scaling experiment is limited to 1, 2, and 4 threads.

The controlled comparisons in Sections 4.3 and 4.4 were run in single-threaded mode by setting

- `OMP_NUM_THREADS=1`
- `OPENBLAS_NUM_THREADS=1`
- `ARMPL_NUM_THREADS=1`

This keeps algorithmic and library-level effects separate from thread scheduling.

B.2 Measurement Protocol

The chapter uses three related measurement modes:

- the benchmark tables in Section 4.2, which report end-to-end wall-clock time together with lightweight stage timers;
- the controlled library-stack and SIMD comparisons in Sections 4.3 and 4.4, which keep the matrix family and threading configuration fixed;

- the deeper profiling and hardware-counter experiments, which use dedicated binaries with heavier instrumentation and are interpreted as explanatory evidence rather than exact timing replicas.

For the benchmark and controlled-comparison tables in Sections 4.2–4.4, each configuration first executes one warm-up run that is discarded, then repeated measured runs with the same wrapper configuration. We use five measured runs for $n \leq 2048$ and three measured runs for $n = 4096$. The tables in the thesis report the median runtime of those measured runs, while the corresponding CSV outputs in the accompanying repository [15] additionally store the mean, sample standard deviation, minimum, maximum, and repeat count. This choice reduces sensitivity to occasional cloud-runtime jitter while preserving the raw variability information for reproducibility.

Timing measurements were obtained using wrapper functions that instrument the main LAPACK stages at the subroutine level. This provides separate timings for DSYTRD, for the tridiagonal eigensolver (DSTEQR or DSTEDC), and for the final eigenvector mapping (DORGTR or DORMTR).

B.3 Additional Profiling Trees

The main text in Section 4.5 reports only the quantitative conclusions from the profiling runs. The detailed inclusive call-tree views are collected here for reference.

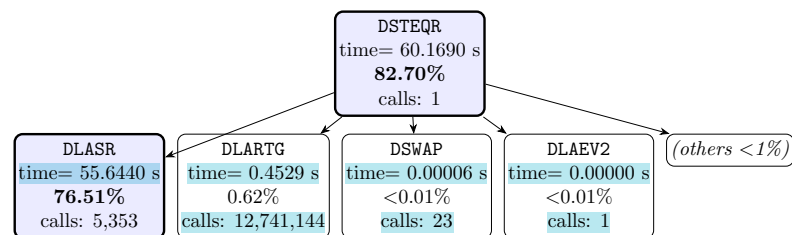


Figure B.1: Hot-path hierarchy inside DSTEQR (inclusive).

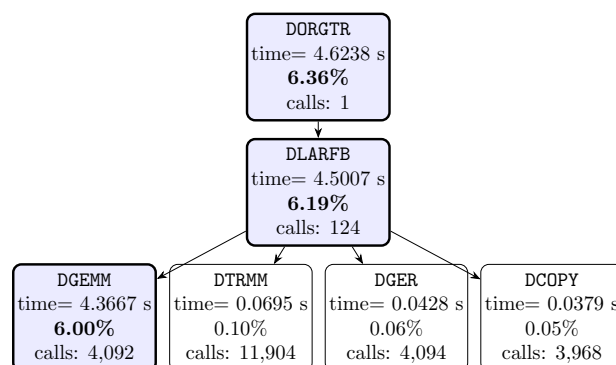


Figure B.2: Hot-path hierarchy inside DORGTR (inclusive).

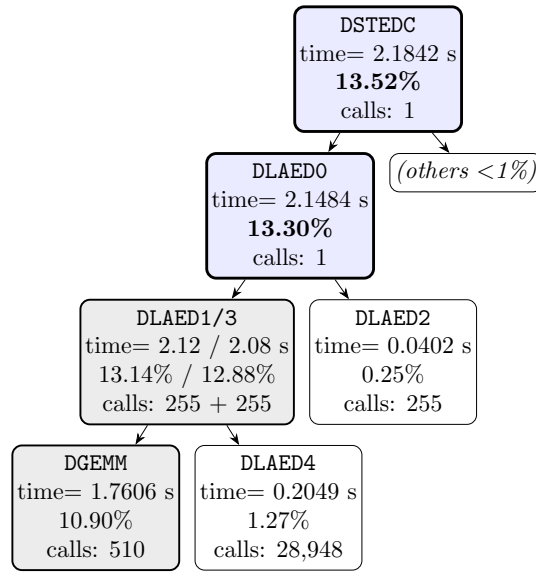


Figure B.3: Hot-path hierarchy inside DSTEDC (inclusive). Node labels show time and relative share.

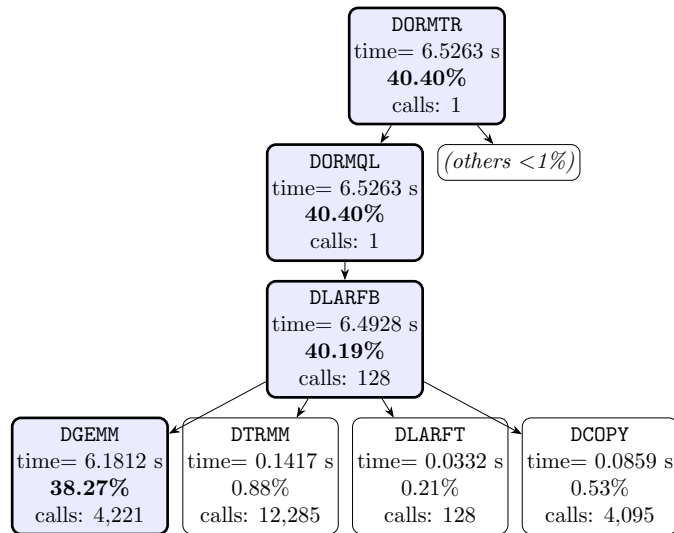


Figure B.4: Hot-path hierarchy inside DORMTR (inclusive). Time and relative share are shown per node.

B.4 Test Matrices and Correctness Checks

All experiments use Symmetric Positive Definite (SPD) Kac–Murdock–Szegő (KMS) matrices with correlation parameter $\rho = 0.95$, defined by

$$A_{ij} = \rho^{|i-j|}, \quad 0 < \rho < 1.$$

This gives a dense symmetric Toeplitz matrix whose off-diagonal entries decay smoothly with distance from the diagonal. The tested matrix sizes are

$$n \in \{512, 1024, 2048, 4096\}.$$

For illustration, when $n = 5$ and $\rho = 0.95$,

$$\begin{bmatrix} 1 & 0.95 & 0.9025 & 0.8574 & 0.8145 \\ 0.95 & 1 & 0.95 & 0.9025 & 0.8574 \\ 0.9025 & 0.95 & 1 & 0.95 & 0.9025 \\ 0.8574 & 0.9025 & 0.95 & 1 & 0.95 \\ 0.8145 & 0.8574 & 0.9025 & 0.95 & 1 \end{bmatrix}.$$

Numerical correctness was verified by computing the standard LAPACK residuals

$$\frac{\|AV - V\Lambda\|_F}{n\|A\|_F}, \quad \|V^\top V - I\|_F,$$

which remained within recommended thresholds for all reported experiments.

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